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Predicting resistance of bearing steels to surface initiated fatigue: Application to hybrid contact

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ABSTRACT

The constant demand for improved energy efficiency in lubricated rolling/ sliding contacts, such as in rolling bearings, is generally addressed through increased power density and the use of low-viscosity (i.e., thin-film) lubricants. In these conditions, mixed lubrication regime can dominate with intensive asperity interactions between two contacting surfaces, thus increasing the risk of surface-initiated fatigue (e.g., in the form of micropitting). In the present study, the influence of steel mechanical properties on the micropitting accumulation in a hybrid contact (steel versus Si_3N_4 ceramics of different roughness) has been experimentally and theoretically investigated. It was found that higher roughness of Si_3N_4 surface could increase the risk of micropitting on the steel component. However, despite formation of some mild micropitting, the bearing steels tested here show good performance. For better understanding of the relationship between the mechanical properties of bearing steels and micropitting resistance, a new theoretical framework is introduced. The new micropitting model, based on the finding that the nucleation of a micropit is driven by (i) local accumulation of the plastic strain in mixed rough contact, and (ii) reduction of the total energy by debonding of the plastically deformed zone is introduced. The new micropitting model was shown to match with experiments across the range of test conditions. Micropitting performance maps based on the mechanical properties of steels (yield strength, work hardening and fracture toughness) have been obtained. The new theory has implications not only for predicting micropitting but also for predicting adhesive wear resistance of different steels.

1. Introduction

Surface initiated fatigue is a failure mode of rolling contact metallic surfaces which occurs in poor lubrication regimes and when certain amount of sliding motion is also present. Due to its practical importance across various industries, different terms such as *surface distress*, *grey staining*, and *micropitting* have been used for describing surface initiated fatigue [1]. For the sake of simplicity we refer to surface initiated fatigue as micropitting throughout the manuscript. Micropitting is usually manifested by ~ μm size (i) cracks and (ii) spalls which are perpendicularly oriented to the rolling–sliding direction (see Fig. 1). Specific orientation of the micropitting with respect to the rolling direction indicates that the surface tractions due to asperity interactions of the two contacting surfaces drive their formation [2]. Micropitting is not one of the primary failure modes in bearing applications. However, formation of micropits leads to even higher surface stresses, which might lead to the reduction in the lubricant film thickness, more pronounced vibration signal [3] and eventually more severe damage such as, but not limited to, surface initiated spalling.

Micropitting has been studied intensively be means of experiment and modeling [2,4–13]. The following micropitting controlling parameters have been identified:

(i) Surface design — topography, and eventual presence of coatings,
(ii) Sliding to rolling ratio S,
(iii) Central lubricant film thickness h_c,
(iv) Functional properties of the surface topography as a function of roughness (e.g. root mean square R_q, skewness R_{sk}, average asperity slope R_{dq}, kurtosis R_k, ...),
(v) Maximum Hertzian contact pressure p_0, and
(vi) Friction and controlled material removal (mild wear) which can be affected by the lubricant chemistry and/or surface coatings.

Essentially, the appropriate control of the above parameters, during both the surface finishing (grinding and/or honing) and bearing exploitation, lead to an improved resistance to micropitting.

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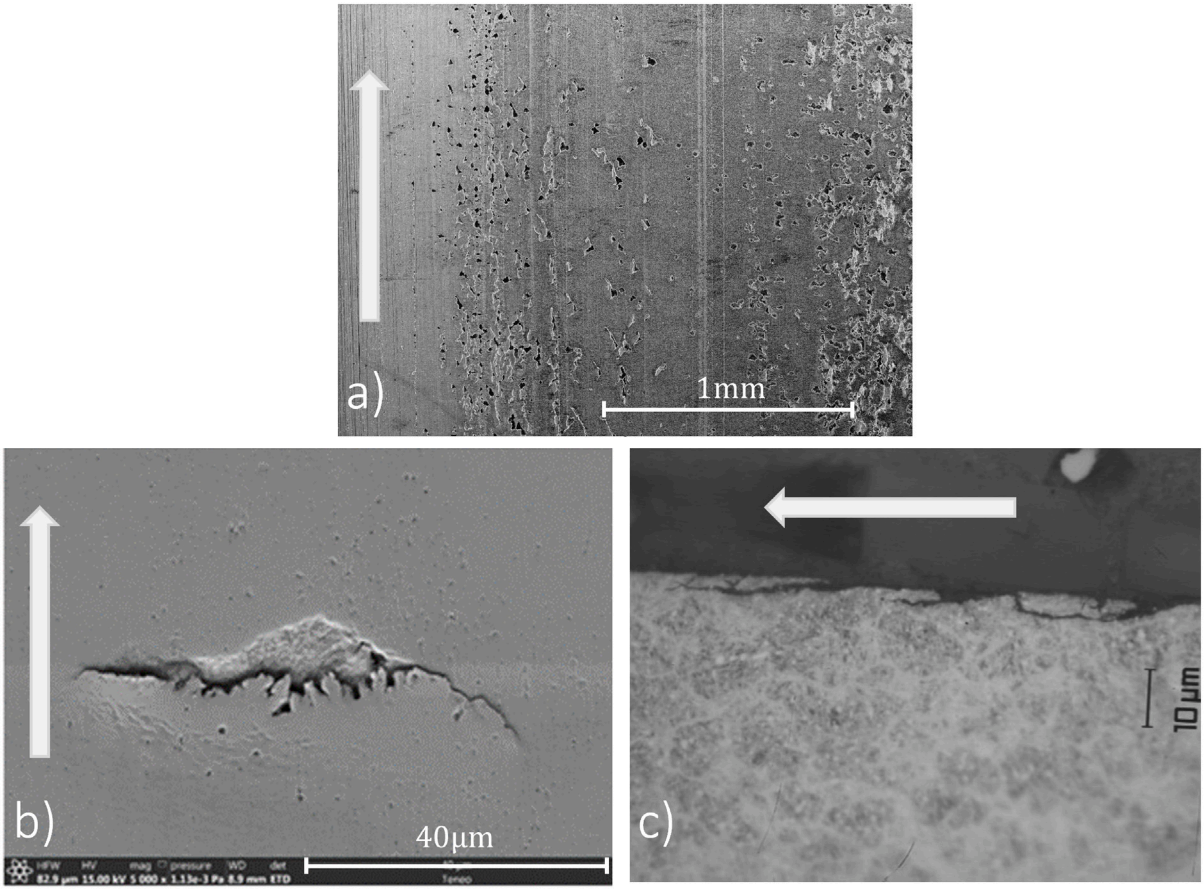

Fig. 1. Scanning electron microscope (SEM) images of (a) micropitted surfaces exposed to rolling–sliding contact. High resolution SEM image of an individual micropit (b) top and (c) side view. The arrow in figures (a) and (b) indicates the rolling direction. Figures (a) and (c) were previously published by the authors in [2].

The above micropitting controlling parameters are not intrinsic to the material of the contacting surfaces. Thus, they can be labeled as *the micropitting extrinsic parameters*. However, micropitting and its manifestation are fundamentally *material failure*. Therefore, mechanical properties of the contacting surfaces play a crucial role in the micropitting resistance of bearing steels. Most of the micropitting related studies in bearings have been performed using the same 100Cr6 martensitic steel, whereas understanding the relationship between different steels and their micropitting performance received less attention. A more systematic approach towards understanding of the micropitting resistance of different bearing steels has been made recently by Sherif et al. [14], and Morales-Espejel and Brizmer [15]. Contrary to common belief, it was found in [14] that there is no strong correlation between the steels hardness and micropitting resistance. It was then postulated that increase in materials 0.2% yield strength should correlate well with the micropitting resistance [14]. However, the exact mechanistic description on how materials stress–strain curve affects the micropitting has not been provided.

Oila and co-workers [16] provided a comprehensive study of microstructural phenomena related to micropitting in martensitic steels. Metallographic examination of fatigued surfaces uncovered several microstructural characteristics absent in untested specimens: near-surface plastically deformed regions (PDR), somewhat deeper dark etching regions (DER), and white etching bands (WEB). The study revealed that the boundaries of PDR can act as sites for the initiation and propagation of micro-cracks, leading to micropitting formation. The finding of Oila et al. [16] is echoed here when a new micropitting fatigue criterion is introduced.

The micropitting risk in bearing applications can be reduced by using so-called "hybrid bearings" [17–19]. Hybrid bearings are manufactured with ceramic rolling elements (typically from Si_3N_4 ceramic) and steel rings. From the contact mechanics point of view, "hybrid contact" represents a contact between the two surfaces which have significantly different chemical, microstructural, mechanical (e.g. hardness, toughness, . . .) and sometimes thermal properties. Detailed rolling bearing fatigue experiments have demonstrated enhanced performance of hybrid bearings in poor lubrication conditions [17]. Enhanced micropitting and adhesive wear resistance of the hybrid contact between Si_3N_4 and 100Cr6 α' has previously been reported by Brizmer et al. [19]. It was found that the improved micropitting resistance of the hybrid contact comes from: (i) reduced roughness of the ceramic surfaces (R_q^{cer} <20 nm, typically observed in ceramic balls), (ii) reduced boundary friction between ceramic and steel in solid-to-solid contact, and (iii) different chemical affinity between the two materials which reduces adhesive forces between the two surfaces [19]. However, as in the case of all-steel contact, effect of the different steel microstructure and mechanical properties on micropitting performance in hybrid contact received less attention. In addition, the effect of changes in ceramic roughness on the micropitting resistance has not been studied up to date.

Here, to address the above mentioned open questions, we perform a systematic study to explore (i) the micropitting resistance of different bearing steels in hybrid contact, and (ii) the effect of ceramic component roughness on the micropitting resistance of steels. We show that the hybrid contact enables very good micropitting resistance of the bearing steels tested here. More specifically, we show that the hybrid contact led to better performance compared to the tests results reported by Sherif and co-workers [14], observed in all-steel contact. Furthermore, we perform a very detailed post-test analysis in which we show the performance differentiation between the steels tested here.

Table 1
The chemical composition of the tested steels. Unless otherwise stated, the concentrations are in wt%.

Steel	C	Si	Mn	P	S	Cr	Mo	W	V	Ni	Cu	Al	N	(ppm) O_2
100Cr6	0.94	0.29	0.41	0.012	0.004	1.47	0.03				0.04	0.09		10
BÖHLER N360	0.31	0.55	0.34	0.016	0.0006	15.11	1.00			0.20			0.40	
PM-1	1.35	0.38	0.32	0.021	0.017	4.53	3.47	0.05	3.60	0.16	0.10			

Table 2
Basic mechanical properties of the three steels tested.

Steel	HV10	±	HRC(150 kgf)	±	$\sigma_{y0.2}$ (GPa)	Q	K_{Ic} (MPa$\sqrt{\text{m}}$)
100Cr6	728.0	3.0	61.6	0.1	2.55	11	17.7
Böhler N360	695.4	2.0	60.3	0.1	2.07	8	17.3
PM-1	767.0	1.0	63.2	0.2	2.71	14	> 20

To gain deeper insights on the fundamental relationship between the steel mechanical properties and micropitting accumulation, we develop a model based on the assumption that the nucleation of a micropit is driven by (i) local accumulation of the plastic strain in mixed rough contact, and (ii) reduction of the total energy by debonding of the plastically deformed zone. The modeling prediction is compared to the experimental measurements and very good agreement is obtained. The new micropitting model captures key aspects responsible for the good micropitting resistance of bearing steels.

The remainder of the manuscript is organized as follows. In Section 2, we give a detailed exposition of the steels used here, the experimental technique for assessing micropitting resistance and final experimental results. In Section 3, we first introduce computational model for simulating elastic–plastic contact between rough surfaces, then we explain the difference in the number of stress cycles experienced by smooth and rough surface in contact, and finally we present an analytical fatigue criteria for predicting the initiation of micropitting. In Section 4, we compare the predictions of the new fatigue criterion for micropitting with the experimental results. In Section 5, we provide a more detailed analysis on the desired properties of steels for good micropitting resistance. Implications of the new fatigue model for micropitting are discussed in Section 6.

2. Experimental performance of advanced bearing steels in hybrid contact

2.1. Steels and heat treatments

The steels investigated in this work are 100Cr6 steel, powder metallurgy (PM-1) steel and BÖHLER N360 ISOEXTRA stainless steel (see Table 1), which were also a part of the test campaign reported in ref [14]. Steel 100Cr6 was heat treated by austenitization followed by oil-quench and tempering. The remaining two steels were vacuum-hardened by austenitization at reduced pressure, followed by gas-quenching. Tempers and subzero treatments were tailored to meet required hardness levels without compromising toughness. A very detailed microstructural characterization of the steel tested here is previously reported in [14], and it is omitted here. The average hardness values after the heat treatment, along with the yield strength measured in compression at 0.2% offset $\sigma_{y0.2}$, hardening exponent Q (see Eq. (3)) and K_{Ic} fracture toughness [14,20] are presented in Table 2 (see Table 1).

2.2. Micropitting tests

To evaluate the micropitting resistance of the steels, the micropitting test rig (MPR), produced by PCS Instruments is employed. The schematic and actual images of the MPR test rig are shown in Fig. 2. Its principal part is the load unit which consists of three rings in contact with the roller manufactured from a steel of interest. To replicate the conditions similar to the hybrid contact (ceramic-steel contact), the three rings are manufactured from Si_3N_4 ceramic. The configuration in which three rings are in contact with one roller leads to fast accumulation of the loading cycles on the test roller. Note that micropitting does not occur on the ceramic surface. The testing is performed in well-controlled oil-lubricated conditions. To increase the likelihood of micropitting as a single failure mode [2], and to replicate the test conditions used in ref [14], we impose slide-to-roll ratio S = 2%. Furthermore, previously reported experimental results suggest that maximizing the roughness differential between the rollers (smooth) and the rings (rough) facilitates accumulation of the micropitting [2] on the roller. To replicate such conditions, the test rollers are polished such that their $R_q^{\text{steel}} \approx 50$ nm while the ceramic rings used in the experiments had different roughness levels ranging between $R_q^{\text{cer}} \approx 80$ nm and 140 nm. To assess the effect of lubrication quality on the micropitting resistance of hybrid contact we vary the central film thickness h_c between the two contacting surfaces. For the sake of simplicity, the lubrication quality is quantified by using simple parameter $\Lambda = h_c/R_{qc}$, where $R_{qc} = \sqrt{(R_q^{\text{steel}})^2 + (R_q^{\text{cer}})^2}$ is composite r.m.s. roughness of the two contacting surfaces. However, the authors are aware that the surface resistance to micropitting is controlled by a set of roughness parameters of the contacting surfaces rather than only by single parameter Λ [13]. Here, depending on the desired conditions, Λ is varied between 0.25 and 0.8. The maximum Hertzian contact pressure is set to 2.5 GPa. VG 32 mineral base oil is used as the lubricant to minimize the influence of tribochemical effects while maximizing the effect of steel mechanical properties on the micropitting [7,19]. The number of over-rolling cycles is set to 5 millions. For better assessment of the (i) nucleation and (ii) growth of individual micropits, the tests were interrupted at N = 0.7, 2 and 5 millions cycles for inspection. Summary of the MPR test conditions is shown in Table 3. Depending on the reproducibility, the micropitting tests are performed at least two times per steel variant. After each test, the rollers tested are visually analyzed for the presence of micro-pits and evaluated according to the scale of micropitting resistance grades, shown in Fig. 3 [14]. The optical images in the SKF micropitting scale (see Fig. 3) are obtained using different steel grades and test conditions described in [14].

Table 3
Micropitting test conditions on the MPR test rig.

Parameter	Value
Max contact pressure p_0	2.5 GPa
Steel roller roughness, R_q^{steel}	≈ 50 nm
Ceramic rings roughness, R_q^{cer}	≈ 80–140 nm
Rings roughness orientation	Isotropic
Λ ratio (Lubrication quality)	0.25–0.80
Slide-to-roll ratio, S	2%
Number of load cycles	0.72, 2, 5 M

2.3. Experimental results

The total micropitted area for all steels tested here against ceramic rings ($R_q^{\text{cer}} \approx 140$ nm) is estimated to be less than 1%. Representative

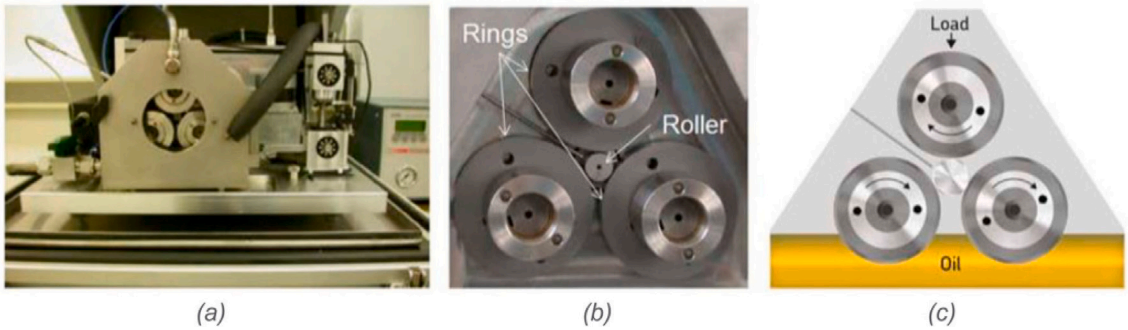

Fig. 2. Micropitting test rig (MPR): (a) general view, (b) load unit and (c) schematic of load application and oil lubrication of the contact.

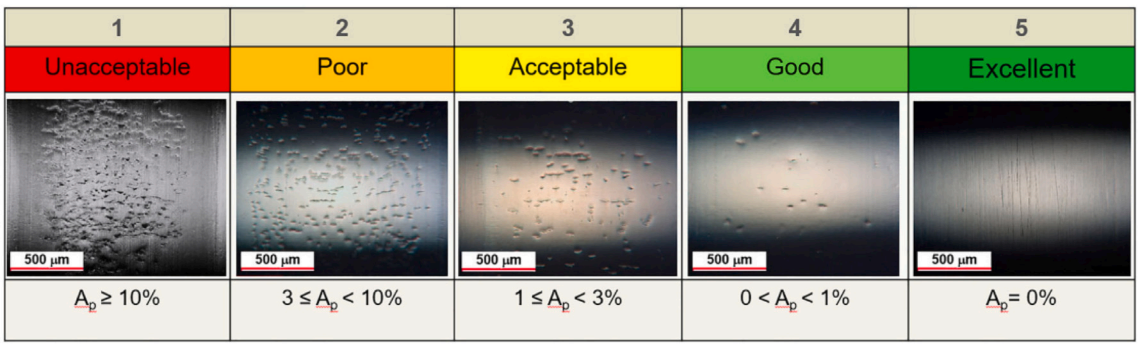

Fig. 3. SKF micropitting resistance scale. A_p denotes the corresponding percentage range of micropitted area [14].

Fig. 4. Micropitted surface of 100Cr6 (left), N360 (middle) and PM-1 (right) steel obtained in micropitting tests after $N = 5$ million cycles at $p_0 = 2.5$ GPa. The ceramics component roughness used in the tests is $R_q^{cer} \approx 140$ nm.

optical microscopy images (see Fig. 4) show seldom appearance of individual micropits after $N = 5$ million cycles and at $p_0 = 2.5$ GPa. Comparing Fig. 4 with SKF micropitting resistance scale (see Fig. 3) we conclude that all the steels tested here can be ranked as either good (100Cr6 and N360) or excellent (PM-1). Compared to the all-steel results reported by Sherif and co-workers [14], we find that the hybrid contact improved the performance of 100Cr6 and N360 steels from "acceptable" to "good" (see Table 12 in Ref. [14]). PM-1 steel shows excellent performance in both "all-steel" and "hybrid" contact. Improved resistance of the hybrid contact has been achieved even though the maximum contact pressure used here is 25% higher and the test duration was 7 times longer as compared to the all-steel test conditions used in [14]. The very good micropitting performance of the hybrid contact is a direct consequence of (i) the reduced adhesion and friction between the steel and ceramic counter surfaces, and (ii) generally lower roughness levels of the ceramics rings as compared to the steel ones used in [14]. However, this finding suggests that the previously known benefits of the hybrid contact are preserved even if different bearing steels are used. This is the first main result of this manuscript.

Even though the above results indicate good micropitting resistance of the steels tested, it does not provide insights about the specific differences between them, which do exist. More specifically, since the well-established micropitting scale (see Fig. 3) is defined for all-steel contact, we cannot provide details about the performance differentiation between the steels in hybrid contact. Therefore, to assess the steel performance differentiation with respect to their micropitting resistance in hybrid contact, we analyze the following additional aspects as a function of the number of cycles (over-rolling):

(i) *Number* of nucleated micropits,
(ii) Distribution of the micropits *width*, and
(iii) Distribution of the micropits *length*.

By counting the number of individual micropits, we assess the steel resistance to the nucleation of micron size fatigue crack which is generally driven by (i) friction tractions due to asperity interactions on the surface, and (ii) by the (tensile) residual stresses due to accumulation of the plastic strain around the asperity summits (see Section 3.4). Furthermore, by measuring the changes in the width and length of individual micropits, we aim to assess the differences in fatigue crack-growth resistance of the steels, which is typically dominated by Mode II. The number of micropits and their size (width and length) has been quantified by inspection of the whole running track of the test specimens (steel rollers) using a microscope. The above quantitative exercise is repeated for every tested surface. Note that for a fixed test conditions (contact, pressure, film thickness and surface roughness) a very good test reproducibility was achieved.

Fig. 5(a) shows that PM-1 is the most resistant steel to the nucleation of new micropits, followed by 100Cr6, while the biggest number of newly formed micropits is found in N360. Using the same data-set, we also find that the steel ranking with respect to the nucleation rate is also the same. However, it is interesting to notice that nucleation rate in N360 slows down with time and thus suggesting an increased resistance to micropitting nucleation in later stages of the MPR tests. Next, Fig. 5(b) shows the box plot distribution of the micropits width after $N = 5 \cdot 10^6$ over-rollings. Again, the best performing steel is PM-1, followed by N360 and 100Cr6, respectively, for which the median value of N360 is slightly lower than 100Cr6. However, despite slightly better performance of N360, we do notice larger number of outliers as compared to 100Cr6. Finally, Fig. 5(c) shows the box plot distribution of the micropits length after $N = 5 \cdot 10^6$ over-rollings. As in the two previous cases, the best performing steel is PM-1, followed by 100Cr6 and N360. Again, the bottom two steels shows similar median performance. However, as in the previous case, N360 shows an increased number of outliers with their length much larger than the median value. This behavior might be a consequence of the larger local fluctuations of the toughness in N360 which can be related to the statistical distribution of carbides (see Figure 6(b) in Ref. [14]). Using the above aspects we derive the final steel ranking with respect to their micropitting resistance in hybrid contact as follows: (i) PM-1, (ii) 100Cr6 and (iii) N360. This is the second main results of this manuscript. Interestingly, the final steel ranking is very similar to that obtained in all-steel contact [14]. Thus, we conclude that hybrid contact improves the micropitting resistance while it does not seem to affect the final ranking of the steels tested here and in [14].

The high roughness levels of the Si_3N_4 surface used in the tests above ($R_q^{cer} \leq 140\,\text{nm}$) are not typical in bearing applications, neither for ball nor for roller bearings. Thus, to mimic the test conditions typical for hybrid roller bearing applications, we perform additional micropitting tests against smoother Si_3N_4 surface ($R_q^{cer} \approx 80\,\text{nm}$) and N360 steel. Note that this steel is specifically chosen since it showed tendency for the easiest nucleation of individual micropits against rough Si_3N_4 surface. Fig. 6 shows the tested surfaces of N360 steel after (a) $N = 0.7 \cdot 10^6$, (b) $N = 2 \cdot 10^6$ and (c)$N = 5 \cdot 10^6$ stress cycles. As expected, we do not find presence of micropits even though the tests were carried out at $p_0 = 2.5\,\text{GPa}$ and $\Lambda = 0.25$. In addition, the presence of numerous dents did not lead to the formation of micropitting, even though it is known that such features can act as the stress-concentration sites [21]. This finding indicates that the other two steel variants would perform equally good against smoother Si_3N_4 surface. More importantly, this result confirms that N360 steel is expected to show an extremely good performance in hybrid ball bearings which are known to have very fine roughness.

Finally, we perform additional micropitting tests with N360 steel against very rough Si_3N_4 with $R_q^{cer} \approx 160\,\text{nm}$ and at different Λ. Fig. 7 shows number of nucleated micropits versus the number of over-rollings for $\Lambda = 0.5$ (dashed line) and $\Lambda = 0.8$ (solid line). Improving the lubrication conditions to $\Lambda = 0.8$ completely prevented the nucleation of micropits after $N = 5 \cdot 10^6$ over-rollings. On contrary, reduced lubrication quality ($\Lambda = 0.5$) led to micropitting as early as 0.7millions of over-rollings. While not shown here, similar performance is obtained with 100Cr6 steel, while PM-1 did not show any signs of micropitting. Therefore, the above results clearly suggest that increased micropitting risk due to rough Si_3N_4 surface can be completely mitigated by increasing the lubricant film thickness and Λ ratio to 0.8. Similar results have been obtained by Wainwright et al. [13]. The mechanistic reasons for the behavior observed are discussed in the following section.

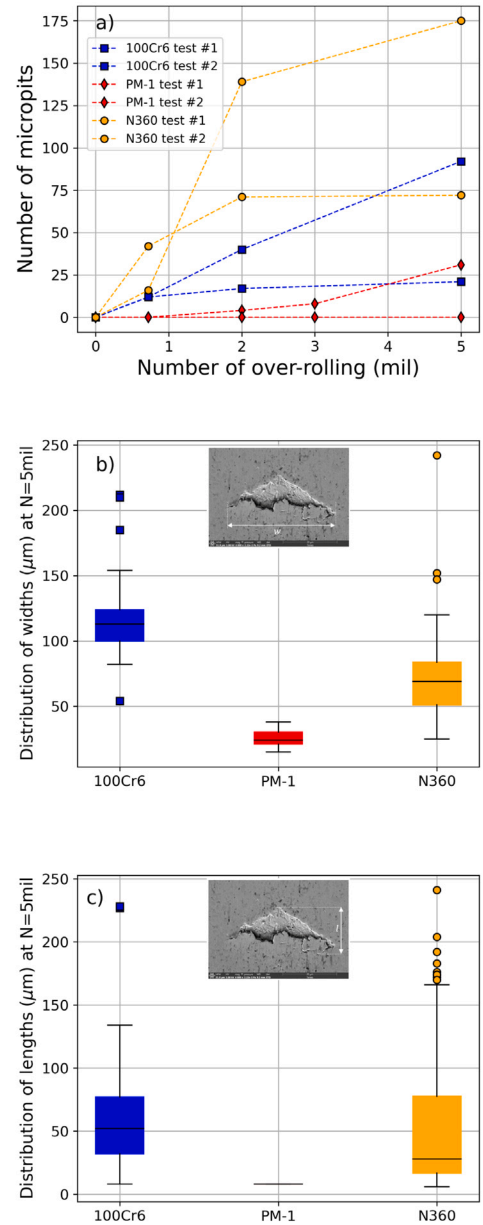

Fig. 5. Micropitting performance of 100Cr6 (blue squares), PM-1 (red diamonds) and N360 (orange circles) steels. (a) Number of nucleated micropits versus number of over-rolling (cycles) measured after first (#1) and second test (#2), (b) Box plot of distribution of micropit widths after 5 million over-rollings and (c) Box plot of distribution on micropit lengths after 5 million over-rollings.

3. Mechanics of micropitting: Computational model and micropitting criterion

Micropitting (MPR) tests are shown to be an efficient experimental method for steel screening with respect to their micropitting resistance. However, due to time and cost constraints, MPR tests are limited to a narrow set of operating conditions, and extrapolating the test results to a wider range of conditions is not a trivial task. Furthermore, MPR tests do not give deep insight into the physical mechanisms governing the micropitting performance of different steels. To strengthen our predictive capabilities beyond MPR tests, we now seek for a physics based mechanistic model which accurately describes and predicts the micropitting resistance of *different bearing steels*. Even though the focus of this manuscript is on steels performance in hybrid contact,

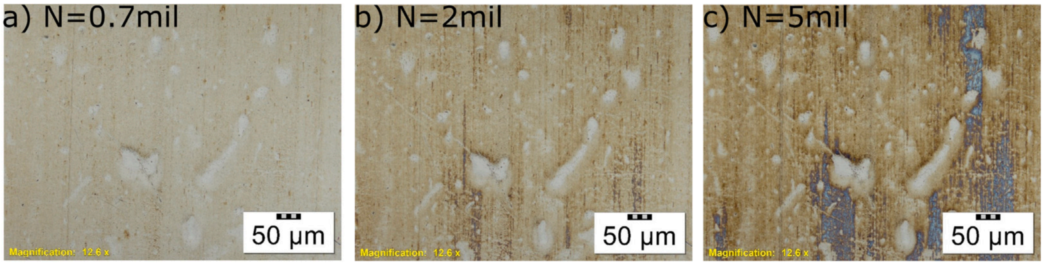

Fig. 6. Tested surface of N360 steel after (a) $N = 0.7 \cdot 10^6$, (b) $N = 2 \cdot 10^6$ and (c)$N = 5 \cdot 10^6$ stress cycles against smooth Si_3N_4 surface (R_q^{cer} ≈80 nm). The tests were performed under p_0 =2.5 GPa and $\Lambda = 0.25$.

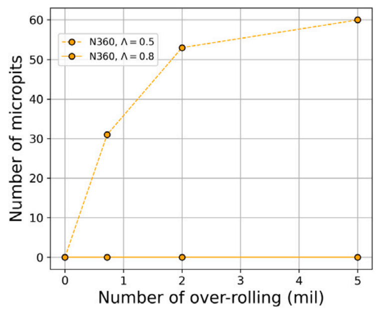

Fig. 7. Number of nucleated micropits versus the number of over-rollings obtained with N360 steel against rough Si_3N_4 surface (R_q^{cer} ≈160 nm) at p_0 =2.5 GPa and $\Lambda = 0.5$ (dashed line) or $\Lambda = 0.8$ (solid line).

the analysis presented below is general and can be applied to both all-steel [2,6,8] and hybrid contact [19].

3.1. Computational model for simulating partly lubricated contact between elasto-plastic rough surfaces

Micropitting failure mode appears when lubrication conditions do not prevent contact between asperities of the two contact surfaces. Such conditions are usually labeled as “Mixed contact problem” or ”Partly lubricated contact" between rough surfaces. To simulate such conditions we employ the model of Morales-Espejel et al. [22]. Here, we only outline the main features of the model while the detailed mathematical description can be found in Refs. [7,22].

The computational model of Morales-Espejel and co-workers solves the contact problem between two periodic, and nominally flat and rough surfaces in rolling–sliding motion. The surface and subsurface coordinates are defined via the Cartesian coordinate system with the axes labeled as x, y and z. The rolling–sliding direction is along x coordinate, the transverse direction is along y and the distance from the surface is measured along z. The simulated surface topographies are 3D replicas of the surface topographies used in the tests and their functional properties are described using the classical tribological parameters (e.g. root mean square, skewness, kurtosis, …). Next, consider that the applied load (contact pressure), lubricant rheology and viscosity, friction coefficients for solid-to-solid and lubricated contact, material constitutive behavior and percentage of the slide-to-roll ratio are known and defined as input parameters. The mean film thickness and mean pressure are also assumed to be known, which is a fairly good approximation when the contact problem of interest is positioned in the center of a macroscopic contact. This approximation is based on the fact that the mean film thickness and mean pressure depend on the macroscopic contact geometry. Therefore, we exclude specific geometrical features which are not relevant for the overall understanding of the mechanistic aspects of surface fatigue and formation of micropits.

Solution to the above boundary value problem are clearance $h(x, y)$, surface displacements $u_i(x, y) = [u_x(x, y), u_y(x, y), u_z(x, y)]^T$ and surface traction $t_i(x, y) = \sigma_{ij} n_j = [t_x(x, y), t_y(x, y), p(x, y)]^T$ where entries of $t_i(x, y)$ are surface pressure $p(x, y) = t_z(x, y)$ and friction tractions $t_x(x, y)$ and $t_y(x, y)$. Surface pressure and displacement profiles in dry contact are solved using FFT based method [23]. The pressure profile in the lubricated area is computed using so-called rapid approach [22]. The surface friction tractions are calculated as

$$t_i(x, y) = f_i p(x, y), \quad i = x, y \tag{1}$$

where

$$f_i = \begin{cases} f_i^{dry} & \text{(for dry contact)} \\ f_i^{lub} & \text{(for lubricated contact)} \end{cases} \tag{2}$$

are friction coefficients for dry and lubricated contact, respectively, and given as input data. Finally, the sharing load between asperities and lubricant is computed using the classical iterative method of flow balance as initially described by Johnson [24]. Once the complete solution for the surface EHL tractions is obtained, the solver calculates the asperity stresses and strains. Due to small physical size of the stressed material in the rough and partly lubricated rolling–sliding contact the application of classical iterative methods such as Finite Elements or Finite Differences comes with the large computational cost. Therefore, the subsurface stress state is computed using an extremely efficient FFT and superposition based method [22], or boundary integral method [25].

Zones of solid-to-solid contact in partly lubricated rough contact can experience local pressure $p(x, y)$ several times higher than the external p_0 [2], but should nowhere exceed the maximum value p_{trunc}. Extremely high surface pressure in these zones inevitably leads to the plastic deformation around asperity summits, and consequently the nonlinear relaxation of the surface pressure as compared to the idealized elastic solution. The actual distribution of the surface pressure and the subsurface plastic strain depend on the material elastic–plastic constitutive behavior. An efficient computational model to solve dry elastic–plastic contact between rough surfaces has been developed for elastic-perfectly plastic material constitutive behavior [26–29]. Recently, Molinari and co-workers introduced a very efficient boundary integral based computational model for elastic-linear hardening materials [25].

However, typical stress–strain curve of a bearing steel shows pronounced nonlinear hardening after the onset of yielding, and thus, solving the contact problem by assuming either linearly elastic-perfectly plastic or linearly elastic-linearly plastic constitutive behavior does not capture the typical surface stress state in bearing applications. To the best of our knowledge, there is no available computational method with acceptable computational efficiency needed in engineering applications and capable to simulate the elastic–plastic behavior typical for bearing steels. In addition, most of the available elastic–plastic rough contact solvers do not take lubrication effects into account; surface tractions are calculated only for dry contact. We thus look for an approximate method capable to reasonably well capture the steel effect on the distribution of surface tractions in bearing applications with partly lubricated rough contact.

3.2. Approximate elastic–plastic model for partly lubricated rough contact

An approximate solution for the distribution of surface pressure and displacement in elastic-perfectly plastic contact has been used previously by the current authors [2]. A similar approach has been adopted here and thus we skip the detailed derivation of the constitutive equations (see Ref. [2] for more details). Here, we only discuss the modifications introduced for better differentiation of bearing steels on their micropitting performance.

First, to account for the plastic behavior of the contacting surfaces we follow the scheme of Tian and Bushan [28] in which the plastic deformations are confined to a small volume (usually around asperity summits) and surrounded by the linearly elastic material. Such a condition enables that the material of contacting surfaces satisfies the condition of "small scale yielding". Next, a contact zone starts to deform plastically once the local contact pressure exceed the material yield strength. The magnitude of the contact pressure in the plastically deformed zones depends on the material constitutive behavior. A typical approximation, previously used by the present authors assumes that the material constitutive behavior can be approximated as an elastic-perfectly plastic and thus leading to a condition that $p_{max} = p_{trunc} = 3\sigma_y$, where p_{trunc} is the limiting pressure and σ_y is the material yield strength. However, a typical bearing steel shows significant work hardening after the initial yielding and therefore, the elastic-perfectly plastic approximation is generally not suitable. Strictly speaking, the contact pressure in the zones of plastic deformations is $p_{trunc} > 3\sigma_y$ [30]. The actual coefficient which relates p_{trunc} and σ_y depends on the materials hardening behavior. We now introduce an approximate method to account for the steel hardening behavior which also satisfies a certain level of simplicity needed in engineering applications.

Assume that the steel constitutive behavior can be described using Ramberg–Osgood power-law equation

$$\varepsilon = \frac{\sigma}{E} + 0.002\left(\frac{\sigma}{\sigma_{y0.2}}\right)^Q \tag{3}$$

where $\varepsilon = \varepsilon_e + \varepsilon_p$ is the total strain composed of elastic ε_e and plastic ε_p constituents, Q is the material hardening exponent, $\sigma_{y0.2}$ is the material yield strength at 0.2% offset, E is Young's modulus and σ is the applied stress. Note that higher value of Q leads to smaller work hardening after the yield point and vice versa. The above equation can be inverted such that it corresponds to strain/displacement boundary conditions as often envisioned in contact problems.

Using the latter approach and for the total strain of $\varepsilon \approx 10\%$ [31] in which $\varepsilon = \varepsilon_e + \varepsilon_p$ and $\varepsilon_p \gg \varepsilon_e$, we estimate what the value of $\sigma = f(\varepsilon)$ is. The value for $\varepsilon \approx 10\%$ is rather arbitrary but reasonable to expect locally due to metal-to-metal contact in mixed lubrication regime [31]. Finally, the truncation pressure, which accounts for the steel hardening behavior, can thus be defined as:

$$p_{\text{trunc}} = c_{\text{trunc}}\sigma(\varepsilon = 0.1) \tag{4}$$

where c_{trunc} is a proportionality constant. $c_{\text{trunc}} = 1.61$ when $\sigma = \sigma_{vm}$ (von Mises stress) and $\nu = 0.3$ at the point of yield inception and when the pressure distribution corresponds to that of Hertz theory [24]. However, the solid-to-solid contact between rough surfaces leads to both (i) large plastic deformation, and (ii) material hardening. Thus, c_{trunc} is expected to have a higher value. Here, we set $c_{\text{trunc}} = 3$ which enables convergence towards the elastic-perfectly plastic p_{trunc} with reduction of the material strain-hardening capability or $Q \to \infty$. With the identified p_{trunc} we now employ the model previously introduced by Morales-Espejel and co-workers as follows:

(i) Perform elastic calculations of surface pressure distribution as described in [2]
(ii) Truncate the elastic pressures to p_{trunc} which is a function of the material constitutive behavior
(iii) Calculate $\delta p(x,\ y) = p(x,\ y)' - p(x,\ y)$ where $p(x,\ y)$ is the pressure distribution from the previous iteration
(iv) Check for convergence using $\text{mean}\{\delta p(x,\ y)\}/p_{\text{target}} = \text{error}$ and repeat until convergence, and
(v) Truncate $p'(x,\ y) < 0$ such that all constrains in the rough contact are met.

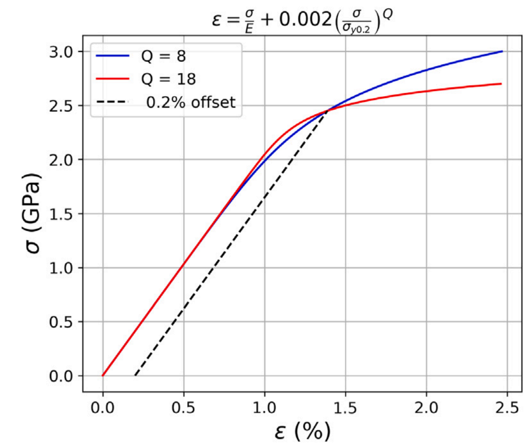

Fig. 8. Example of two model steels having identical 0.2% yield strength but different hardening exponent Q. The material with lower work hardening (bigger Q) will develop smaller values of contact pressure in solid-to-solid contact as compared with the material with higher work hardening (lower Q).

The plastic displacements are calculated by subtracting the elastic part from the total displacements, while the total displacements are calculated using the method described by Stanley and Kato [23].

To demonstrate the importance of including both the material yield strength $\sigma_{y0.2\%}$ and work hardening exponent Q in estimating the maximum surface pressures in partly lubricated rough contact, we now employ the model described above. More specifically, we compare two contact cases in which surfaces in contact have identical $\sigma_{y0.2\%}$ =2.45 GPa but different hardening exponent Q, as shown in Fig. 8. As for the extrinsic contact parameters, we assume contact between two nominally flat rough surfaces; the upper surface is having 15 time bigger roughness $R_{q2} = 15R_{q1} = 150$ nm. The lubricant considered in the simulation is mineral oil VG 32 with dynamic viscosity $\eta_0 = 9.4 \cdot 10^{-3}$ Pa · s at 75 °C. The piezo-viscosity coefficient is assumed to be $\alpha = 20.78$ GPa^{-1}. The solution presented below is static in nature with maximum Hertzian contact pressure $p_0 = 1.5$ GPa. We set the central film thickness h_c such that the ratio $\Lambda = h_c/R_{qx} \approx 0.7$. The slide-to-roll ratio and the lubricant entrainment velocity are chosen to be $S = 2\%$ and of $U_x = 1.146$ m/s, respectively. We do not consider sliding in y direction. Material Young's modulus and Poisson's ratio are $E = 206$ GPa and $\nu = 0.3$, respectively. For the sake of clarity, the above EHL modeling parameters are listed in Table 4.

Using the above defined contact conditions and Eq. (4), we calculate the maximum value of the local pressure for the two model materials with hardening exponents $Q_1 = 8$ and $Q_2 = 18$ to be $p^1_{\text{trunc}} = 11.7p_0$ and $p^2_{\text{trunc}} = 9.1p_0$, respectively. The estimated values correspond to the experimentally measured average microhardness of typical bearing steels (see Figures 9 and 11 in Ref. [14]). In addition, since the initial yield strength of the two model materials is approximately $\sigma_y^{\text{ini}} \sim 1.8$ GPa (see Fig. 8), the ratio $p_{trunc}/\sigma_y^{\text{ini}}$ is 6.5 and 5, for materials 1 and 2, respectively. These values are in general agreement with those obtained using finite element simulations for sinusoidal [32,33] and rough surfaces [34]. As indicated earlier, the increased work hardening of a bearing steel will inevitably result in higher local contact pressure in partially lubricated rough contact. Estimating a value for pressure truncation by using only $\sigma_{y0.2}$ can lead to a wrong estimate of p_{trunc}. Implications for micropitting resistance as a function of the pressure truncation is discussed later in the text.

Table 4
EHL contact parameters used in the Mixed Lubrication model.

R_{q1} (nm)	R_{q2} (nm)	η_0 (75 °C) (mPa s)	α (GPa^{-1})	S –	U_x (m/s)	U_y (m/s)	E_{steel} (GPa)	ν_{steel} –	E_{cer} (GPa)	ν_{cer} –
10	150	9.4	20.78	2%	1	0	206	0.3	305	0.28

3.3. Difference between macro and micro stress cycles in rolling–sliding contact

Presence of sliding in rough rolling contact inevitably leads to increased number of the local stress cycles [8]. For every macro cycle (over rolling) there is a number of stress microcycles at the asperity level. To quantify the actual number of the microcycles for every over rolling we employ the Matlab implementation[1] of the so-called "rainflow counting algorithm" [35] in the following way. First, we track the magnitude of the surface pressure fluctuations during one over-rolling (as a function of time) for every asperity in contact. Essentially, every contacting asperity experiences different pressure history due to statistical nature of mixed lubrication regime. Then, with the identified pressure history for every surface point, we calculate what the magnitude and total number of pressure fluctuations are during one over-rolling. Only cycles for which the local pressure exceeds 1.5 GPa [36,37] are included since we assume that the lower pressure does not lead to fatigue. Note that the local pressure is not the same as the value of Hertzian contact pressure. Then, by applying the rainflow counting algorithm to the pressure history during one over-rolling we extract what the local number of stress cycles is. Finally, the total number of cycles per surface point is

$$N_T(x,\ y) = N_M n_\mu(x,\ y) \tag{5}$$

where $n_\mu(x,\ y)$ is the local number of microcycles during one over-rolling, N_M is the total number of macro stress cycles which is measured in experiments/application (number of over-rollings), and $N_T(x,\ y)$ is the total number of cycles which is the local surface quantity.

Fig. 9 shows simulated $n_\mu(x,\ y)$ for two model surfaces in contact and having different roughness levels, namely R_q =10 nm (left figure) and R_q =150 nm (right figure). In addition, the rougher surface is transversely oriented with respect to the rolling/sliding direction. The algorithm proposed predicts that the smoother surface experiences much bigger number of the local stress cycles, compared to the rough surface. Thus, it is not surprise, that the smooth surface is more prone to micropitting when in contact with the rough surface [2]. Here, we also take the opportunity to point out that even though MPR tests can be regarded as short (1–5 million macro stress cycles), the actual total number of cycles experienced by the surface material can sometimes be up to three times bigger.

The above analysis provides all the necessary input for fatigue calculations. The summary of the complete computational model is presented in Fig. 10. In the following section we present fatigue criterion for predicting micropitting in rolling–sliding partly lubricated rough contact.

3.4. Localized plasticity micropitting criterion (LPMC)

Localized plasticity micropitting criterion (LPMC) is based on the previous work of Kato, Mori and Mura [38]. The analysis presented here is general and not limited to hybrid contact. The statistical nature of the rough contact (see Fig. 11(a)) leads to the localization of the plastic strain around the asperity summits. Fig. 11(b) shows an example of the localized plastic strain after two nominally flat and rough surface came into contact under an externally applied contact pressure p_0 =1.5 GPa. The calculation details are provided in the caption of Fig. 11. Furthermore, the zones of the localized plasticity will essentially be separated by the linearly elastic material. Thus, the localized zones of the plastic strain can be envisioned as the materials "inclusions" which undergo eigenstrain equal to the inelastic plastic strain $\varepsilon^T_{ij} = \varepsilon^p_{ij}$ confined within them (see Fig. 11(b)). Note that the term "inclusion" used here is not a non-metallic inclusion, but a mathematical formulation needed for describing the localized plastic strain. Since plasticity leads to the relaxation of the local stresses during the contact between surface asperities (solid–solid), the material fatigue essentially occurs during unloading when locally plastically distorted material is constrained by the surrounding elastic material and thus leading to the build-up of residual (tensile) stress.

Presence of eigenstrain, which is surrounded by a linearly elastic material, will lead to the increase of the system total energy. We point out here that the energy increase is not related to the dissipated energy such as heat losses due to friction. Neglecting the interaction between the zones of localized plasticity, the energy increase upon unloading (after first stress cycle) can be calculated by using Eshelby theory [39,40] and expressed as

$$E_I = -\frac{1}{2}\int_{V_I} \sigma^I_{ij} : \varepsilon^p_{ij} dV + \Delta E_{half} \tag{6}$$

where σ^I_{ij} is the stress within the plastically deformed zones due to residual plastic strain after the over-rolling (unloading), and V_I is the volume of the plastically deformed region. Note that both σ^I_{ij} and ε^p_{ij} are effectively function of $x,\ y$ and z since the plastic strain depends on the local stress history. The term ΔE_{half} is an additional energy contribution due to half-space effect [41]. Next, cyclic loading will lead to the further increase of the accumulated plastic strain. By assuming that the accumulation of plastic strain follows Coffin-Manson law, the total energy increase after N_T number of cycles is

$$E_I = -\frac{1}{2}\int_{V_I} \sigma^I_{ij} : \Delta\varepsilon^p_{ij}\sqrt{N_T} dV + \Delta E_{half} \tag{7}$$

where $\Delta\varepsilon^p_{ij}$ is the local plastic strain range during one cycle, and N_T is the total number of loading cycles (see Eq. (5)). A micropit nucleates when energy E_I equals the fracture energy required to nucleate a crack of area A (see Fig. 11(c)), or

$$-\frac{1}{2}\int_{V_I} \sigma^I_{ij} : \Delta\varepsilon^p_{ij}\sqrt{N_T} dV + \Delta E_{half} = 2AG \tag{8}$$

where G is the fracture energy. The above condition is equal to the Griffith theory of fracture with E_I being the energy release and driving force for the nucleation of a micropit.

Eq. (8) can further be simplified by assuming $V_I = \mathrm{const}$ and using the homogenization theory

$$-\frac{1}{2}\int_{V_I} \sigma^I_{ij} : \varepsilon^p_{ij} dV \le -\frac{1}{2}\int_{V_I} \bar{\sigma}^I_{ij} : \bar{\varepsilon}^p_{ij} dV \tag{9}$$

where $\bar{\sigma}^I_{ij} = \frac{1}{V_I}\int_{V_I}\sigma^I_{ij}(x,y,z)dV$ and $\bar{\varepsilon}^I_{ij} = \frac{1}{V_I}\int_{V_I}\varepsilon^I_{ij}(x,y,z)dV$ leading to

$$-\frac{1}{2}\bar{\sigma}^I_{ij} : \Delta\bar{\varepsilon}^I_{ij} V_I\sqrt{N_T} + \Delta E_{half} = 2AG. \tag{10}$$

Rearranging Eq. (10) and assuming $\Delta E_{half} \rightarrow 0$ yields the critical number of cycles for the nucleation of a micropit

$$N_C = \left(-\frac{4AG}{\bar{\sigma}^I_{ij} : \Delta\bar{\varepsilon}^I_{ij} V_I}\right)^2. \tag{11}$$

Eq. (11) is general if the small scale yielding condition in satisfied. This is the third main result of the manuscript.

[1] https://nl.mathworks.com/help/signal/ref/rainflow.html

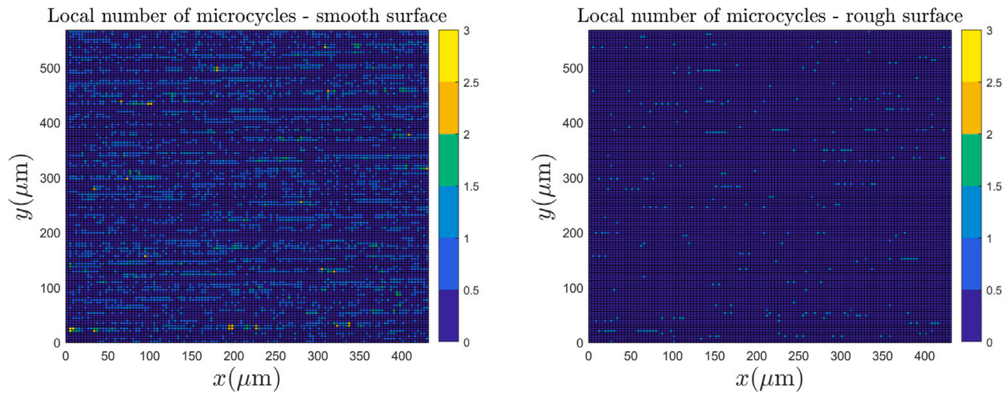

Fig. 9. Distribution of the local number of cycles during one over rolling between smooth (left figure R_q =10 nm) and rough surface (right figure R_q =150 nm). The simulated conditions correspond to p_0 =2 GPa and Λ = 0.25. It is clear that the smooth surface experiences more stress cycles than the rough surface. The simulated rolling direction is along x axes.

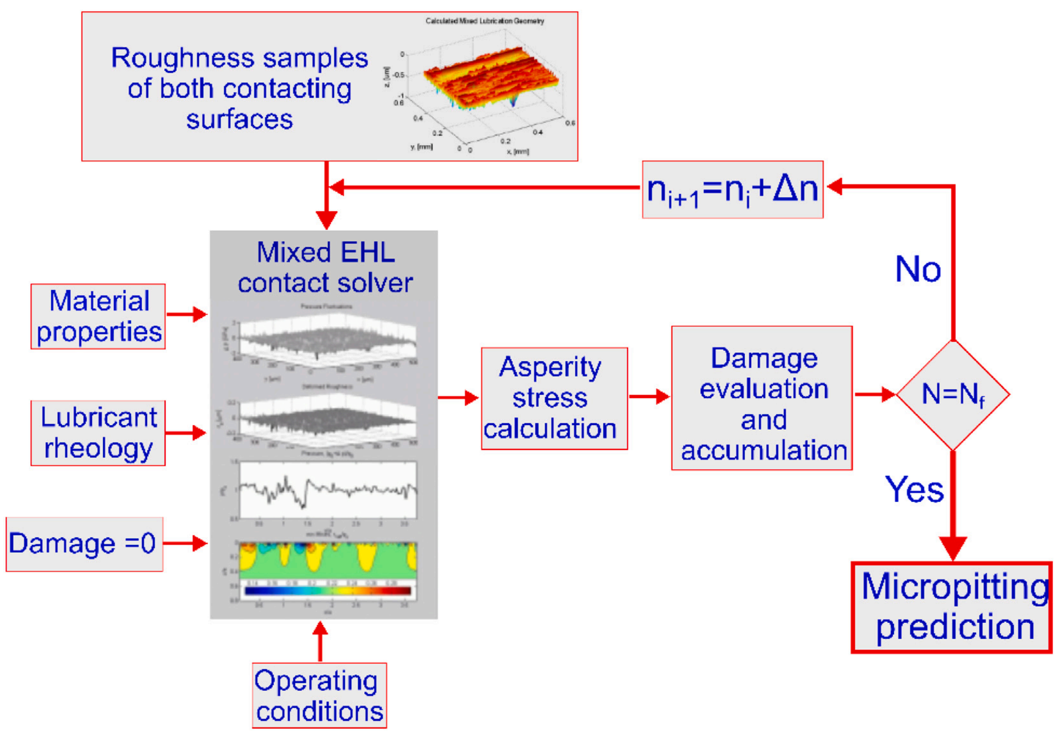

Fig. 10. Flowchart of the full computational model for simulating micropitting.

Next, to find the correlation between the basic steel properties and critical number of stress cycles, the above equation can further be simplified by assuming that (i) the shape of the localized plastic zone can be approximated by a sphere of radius a, (ii) the newly formed crack has circular shape of radius a, and (iii) that the eigenstrain tensor ε^p_{ij} has only shear component which is non zero. The latter approximation is very convenient for fast engineering simulations in which the magnitude of plastic strain is calculated only by means of Glinka method [42] (see A). Then, using the above two approximations yields a closed form analytical solution for calculating the critical number of cycles at which micropit (surface crack) nucleates

$$N_c = \frac{6G}{\mu B \Delta\varepsilon_p^2 a} \tag{12}$$

where μ is the material shear modulus, $B \approx 0.5$ is shape coefficient calculated from Eshelby tensor for sphere, and $\Delta\varepsilon_p$ is the plastic strain range during one cycle. Eq. (12) clearly shows that the materials (i) with higher toughness (larger G) and (ii) which are more resistant to plastic deformations (smaller $\Delta\varepsilon^p$) are expected to exhibit higher resistance to micropitting. The latter condition is generally met if a material has higher hardness. However, since the distribution of the surface pressure is also a function of both $\sigma_{y0.2}$ and Q, it is rather delicate balance between the two. Therefore, using only hardness is not sufficient for the material differentiation, but general trends are expected to hold.

The local surface damage at position $(x,\ y)$ and after the total number of cycles N_T is evaluated as

$$D = \frac{N_c}{N_T}. \tag{13}$$

The condition for crack nucleation is then

$$D \begin{cases} < 1 & \Longrightarrow \quad \text{micropit does not nucleate} \\ \geq 1 & \Longrightarrow \quad \text{micropit nucleates} \end{cases} \tag{14}$$

The fracture energy G is expected to be related to the material fracture toughness. However, it is not immediately clear which value

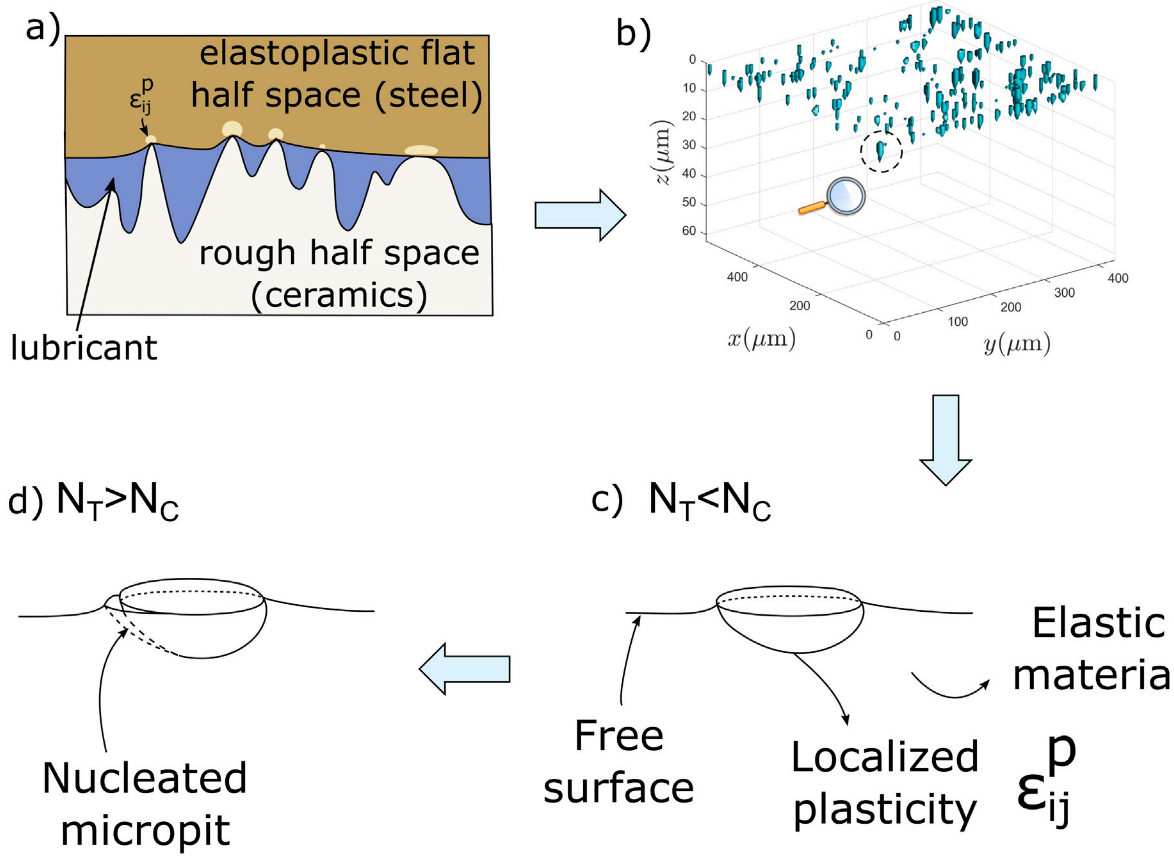

Fig. 11. (a) Schematic representation of a partly lubricated contact between flat elastoplastic half-space (steel) and rough half space (ceramics). Light color indicates localized zones of the plastically deformed metallic surfaces. (b) Estimated zones of the localized plastic strain after nominally flat surfaces (smooth and rough) came into contact under external contact pressure p_0 =1.5 GPa and Λ = 0.7. Note that axes x and y have different scale compared to z axis. (c) Schematic representation of a localized and plastically deformed zone and associated plastic strain ϵ^p_{ij}; criterion for the nucleation of micropit is not met or $N_T < N_C$. (d) Criterion for the nucleation is met $N_T > N_C$ and micropit is nucleated.

to use since the fracture toughness is highly sensitive to the operating length scale. Here, we postulate that G should be proportional to the materials K_{Ic} fracture toughness through

$$K_{Ic} = \sqrt{\frac{2\mu}{1-\nu}G} \tag{15}$$

where ν is material Poisson's ratio, and K_{Ic} is material fracture toughness measured experimentally. Our assumption is based on the observation that individual micropits can be regarded as sharp cracks which open in Mode I loading, and that the size of individual micropits is bigger than the lath/plate size of a bearing steel (see Fig. 1 above and Figure 11 in ref [43]). Opening mode during nucleation of a micropit is driven by (i) tensile stresses on the surface due to friction tractions [24],[2] and (ii) residual *tensile* stresses which arise due to presence of the localized plastic zones upon unloading. We demonstrate point (ii) by plotting the σ_{xx} stresses around the localized plastic zone [39,44] with the residual plastic strain

$$\epsilon^p_{xz} = \begin{cases} 0.01 & \text{if} \quad x^2+y^2 \le 0.5^2 \\ 0 & \text{if} \quad x^2+y^2 > 0.5^2 \end{cases} \tag{16}$$

Fig. 12 shows that even though only shear component of the plastic strain are present within the plastically deformed zone ($x^2+z^2 \le 0.5$) and zero elsewhere, there will be a build-up of the tensile stresses around the plastically deformed zone (within the elastic material) upon unloading. Under rolling contact fatigue, the cyclic build-up of tensile

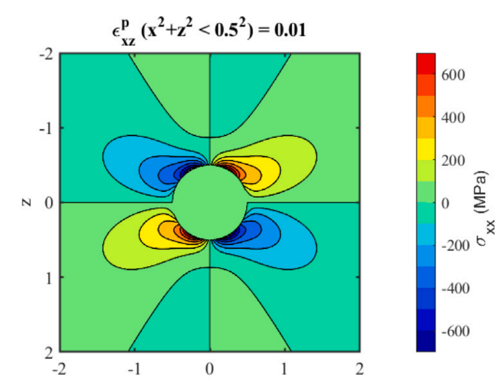

Fig. 12. Residual σ_{xx} stress due to residual plastic strain ϵ^p_{xz} = 0.01 within the localized zone ($x^2+z^2 \le 0.5^2$). The coordinate system is positioned in the center of the plastically deformed zone.

stresses upon unloading coupled to friction stresses will lead to the debonding of the plastically deformed zone, nucleation of a micropit, and Mode I is expected to be the dominant mode. Thus, even though approximate, the condition for using the experimental value of K_{Ic} in calculating G seems reasonable.

In conclusion, the above fatigue model is based on the work of Mura and co-workers [38,41]. However, the main difference between the two is in the observation that a is the size of the localized plasticity while Mura envisioned it to be size of a grain which is oriented in a

[2] For simplified solution one can check Eq.2.31a in Johnson book or it's convolution for more complex traction distributions

way that will lead to the fastest accumulation of the plastic strain and fatigue. In other words, the model introduced here assumes a to be larger than the typical grain size which can be envisioned as a Representative Volume Element (RVE). We also notice that the LPMC shares a couple of important similarities with Continuum Damage Mechanics (CDM) [45–47]. First, both criteria assume the elastic energy release and plastic strain range are the main mechanistic driving forces for the nucleation of fatigue cracks. Second, the positive effect of material's tensile strength (as used in CDM) is effectively taken into account through the material K_{Ic} fracture toughness in LPMC. This relationship holds since the bearing materials with high tensile strength are also known to have higher K_{Ic} fracture toughness [20]. Finally, the above criterion is suitable for predicting the process of nucleation, while predicting propagation of individual micropits requires more advanced computational methods.

4. Model validation

Details of (i) the computational framework for simulating elastic–plastic contact between partly lubricated surfaces (steel–steel or ceramic-steel) and (ii) mechanistic model for predicting micropitting are introduced in the previous section. In this section, we use the above models for predicting the micropitting performance in hybrid contact of different bearing steels. The materials simulated are identical to those used in the MPR tests, namely 100Cr6 α', PM-1 and N360. The summary of their mechanical properties used in the model ($\sigma_{y0,2}$, Q and K_{Ic}) are provided in Table 2. In addition to the steels effect on the micropitting resistance we predict the effect of ceramic surface roughness, lubrication quality and contact pressure. More precisely, the steel surface profiles used in the model are actual replicas of those used in the MPR tests with R_q^{steel} =45.5 nm, 39.1 nm, and 39.9 nm of 100Cr6, PM-1 and N360, respectively. Roughness of the ceramic counter-body used in the model are also replicas of the surface profiles used in the tests for which we vary their R_q^{cer} between 80 and 140 nm. The lubrication quality is analyzed through the parameter $\Lambda = h_c/R_q^c$ where R_q^c is the composite r.m.s. roughness and h_c is the central film thickness. In the simulations, we vary Λ between 0.25 and 1.0. The maximum Hertzian contact pressure was varied between 1.5 and 2.5 GPa. Furthermore, the MPR test results clearly suggested that the roughness of the ceramic component is one of the most important external factors controlling the micropitting performance of a hybrid contact. Therefore, for a more robust analysis of the micropitting performance of hybrid contact, we introduce a new non-dimensional parameter $\chi = R_q^{\text{cer}}/R_q^{\text{steel}}$.

The simulated sliding-to-roll ratio is set to S=2%, as used in the MPR test. The simulated lubricant corresponds to the mineral oil with dynamics viscosity $\eta_0 = 9.4\cdot 10^{-3}$ Pa s at 75 °C. while the piezo-viscosity is taken to be α =20.78 GPa^{-1}. Since we are dealing with mixed lubrication, the friction coefficient is taken separately for lubricated contact $f_{lub} = 0.04$ and for solid-to-solid contact (ceramics-steel) $f_{dry} = 0.05$ [19]. Note that solid-to-solid contact simulated here corresponds to boundary lubrication regime. Thus, the term "dry" used here is only for the easier notation. Related to the friction coefficient, it is also expected that different steels show different friction coefficient in solid-to-solid contact. However, while possible to extract the difference, the measurements were not available to us and so we keep the same dry friction coefficient for all the steels. Finally, unlike in [2], here we do not include effect of the mild wear on micropitting performance. Thus, the results presented here are purely fatigue based.

The micropitting color grades, predicted using LPMC for the three steels under different levels of maximum contact pressure, are shown in Fig. 13. Note that due to statistical nature of the MPR tests, we plot our predictions only as micropitting (color) grade without providing the explicit percentage of the micropitted area. The color grade is set using the previously introduce micropitting resistance scale [14] (see Fig. 3). For every level of the contact pressure p_0 we estimate the steel performance as a function of the lubrication quality Λ and non-dimensional parameter $\chi = R_q^{\text{cer}}/R_q^{\text{steel}}$. Note that additional surface phenomena, such as wear, oxidation, tribo-layer build up, running-in changes in topography . . . which might accompany the micropitting are not included into the numerical model used here. We now analyze the results in more detail and discuss the mechanistic reasons for the performance differentiation between the steels considered here.

Fig. 13 reveals several interesting features. First, LPMC predicts improved performance with increasing Λ and reduction of χ. While the effect of Λ on the micropitting performance is well understood, our analysis reveals that χ plays equally important role in the micropitting performance in hybrid contact. The tribological advantage of the hybrid contact versus all-steel contact disappears in the case when ceramic surface is significantly rougher than the steel counterpart ($\chi \gg 1$). Second, the specific bounds on Λ and χ are steel and contact pressure dependent. Third, the result presented in Fig. 13 suggest that PM-1 is the best performing steel, followed by 100Cr6, while N360 is predicted to be the worst performing steel. Therefore, the model predictions match the experimentally observed steel ranking (resistance to the nucleation of micropits) which was discussed in Section 2.

To further demonstrate the capability of the above mechanistic model for predicting micropitting performance of different steels, we now compare the performance maps simulated for p_0 =2.5 GPa with actual optical microscopy images obtained from the MPR tests (see Figs. 14 to 16). The estimated micropitting risk per steel is generally well captured and reasonably good qualitative agreement is obtained. However, we do notice that LPMC predicts slightly higher micropitted area in N360 than observed in the tests. A possible reason for this quantitative discrepancy requires further investigation.

5. Effect of steel mechanical properties on micropitting resistance: further comments

Using the advanced contact solver for simulating partly lubricated rough contact in combination with the above introduced micropitting criterion, enabled us to predict the surface performance differentiation between 100Cr6, PM-1 and N360 steels. However, despite being rather a powerful method, the above analysis does not provide direct insights on what mechanical properties are desirable for a good micropitting resistance.

To tackle this challenge, we perform an additional modeling campaign in which we simulate micropitting resistance of model steels generated by varying:

(i) yield strength between 1.8 GPa $\leq \sigma_{y0.2} \leq$ 2.8 GPa with the step $\Delta\sigma_{y0.2} = 100$ MPa,
(ii) hardening exponent between $8 \leq Q \leq 18$ with the step $\Delta Q = 1$, and
(iii) fracture toughness between 16 MPa$\sqrt{\text{m}} \leq K_{Ic} \leq$ 30 MPa$\sqrt{\text{m}}$ with the step $\Delta K_{Ic} = 1$ MPa$\sqrt{\text{m}}$

leading to the total of 1815 simulations. The bounds on the material parameters are specifically chosen since they cover the wide range of medium and high-C steels used in bearing applications. Again, we model the rolling–sliding contact between the smooth steel surface (R_q^{steel} ≈50 nm) and rougher Si_3N_4 ceramics (R_q^{steel} ≈140 nm). In all the simulations we fix $\Lambda = 0.3$ and $p_0 = 2.0$ GPa. Effect of mild wear is not included.

Fig. 17(a) shows the simulated percentage of the micropitted area as a function of the material hardening exponent Q and yield strength $\sigma_{y0.2}$. As for the materials fracture toughness, we show results for K_{Ic} = 18 MPa$\sqrt{\text{m}}$ which is a typical value in bearing martensitic hardened and tempered steels [20]. A couple of very important features emerge from Fig. 17(a). First, increasing the material yield strength leads to the reduced micropitting risk. This behavior is a consequence of the reduced steel tendency to plastically deform with increasing $\sigma_{y0.2}$. A similar conclusion, although not based on the mechanistic

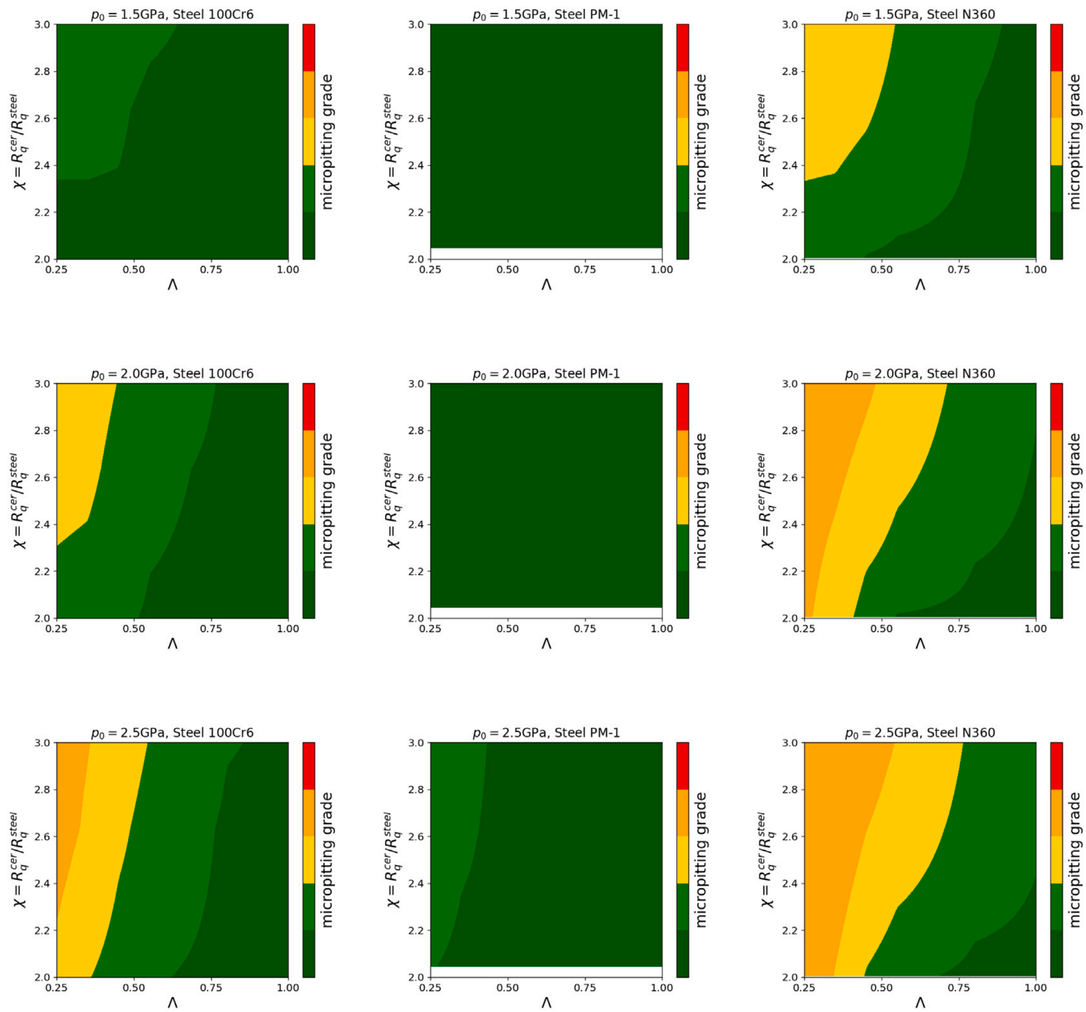

Fig. 13. Micropitting color grades predicted using LPMC in hybrid contact as a function of the lubrication parameter Λ and parameter $\chi = R_q^{\mathrm{cer}}/R_q^{\mathrm{steel}}$ for three different values of the contact pressure $p_0 = 1.5$, 2.0 and 2.5 GPa, and three different steels. Note that the actual χ values used in the tests have wider range than presented in the figure. However, for the sake of easier comparison between the steels, we introduce hard limit on the y-axes values.

modeling, was previously provided by Sherif and co-workers [14]. Second, reduced work hardening capability (larger Q) leads to improved micropitting resistance. Such a behavior is a direct consequence of the reduced contact pressure in solid-to-solid contact which then has positive effects on the micropitting resistance. Our finding is in agreement with the work of Frerot and Molinari (see Figure 4 in Ref. [48]) who used a different solver for modeling dry elastic–plastic contact between rough surfaces. Furthermore, improved fracture toughness with increased yield strength and reduced work hardening capability has been predicted by Tvergaard and Hutchinson [49] who arrived at the similar results by studying a different mechanics problem.

Next, Fig. 17(b) shows the simulated percentage of micropitting as a function of the material fracture toughness K_{Ic} and yield strength $\sigma_{y0.2}$. As for the hardening exponent, we use $Q = 11$ which is typically observed in 100Cr6 steels. Increasing the material fracture toughness leads to the improved micropitting resistance, as expected. However, comparing to the effects of $\sigma_{y0.2}$ and Q, the effect of K_{Ic} on the micropitting seems less pronounced. However, we point out here, that the material properties under consideration cannot be decoupled and high yield strength with lower work hardening is usually associated with higher fracture toughness [49].

Our analysis shows that steels with (i) high yield strength, (ii) low work hardening capability (high Q) and/or (iii) high fracture toughness are expected to have good micropitting resistance in hybrid contact under mixed lubrication regime. This conclusion, along with the performance maps shown in Figs. 17(a) and (b) can be used for estimating the expected micropitting resistance of steels. Using only average hardness for micropitting performance differentiation can lead to inaccurate, and sometimes, wrong predictions. We demonstrate this by plotting the calculated percentage of micropitted area as a function of the materials average microhardness evaluated using Eq. (4) (see Fig. 17(c)); for the same hardness, there can be a factor of ~ 10 difference in the simulated percentage of micropitting which is in qualitative agreement with the experimental results reported by Sherif and co-workers [14]. In addition to the model materials, Fig. 17(c) shows the predicted percentage of micropitting for the three steels studied here, namely 100Cr6, PM-1 and N360. Fig. 17(c) demonstrates that well established bearing steels are generally optimized for a good surface performance in poor lubrication conditions ($\Lambda = 0.3$).

6. Discussion

Hybrid contact between ceramic Si_3N_4 and the three bearing steels tested here has been shown to have very good micropitting resistance across the range of conditions. More importantly, we have demonstrated that the hybrid contact shows superior micropitting resistance even with increased roughness level of the ceramics surfaces ($R_q \approx 140$ nm). Reducing the ceramic roughness level, as typically observed in bearing products, further mitigates the micropitting risk.

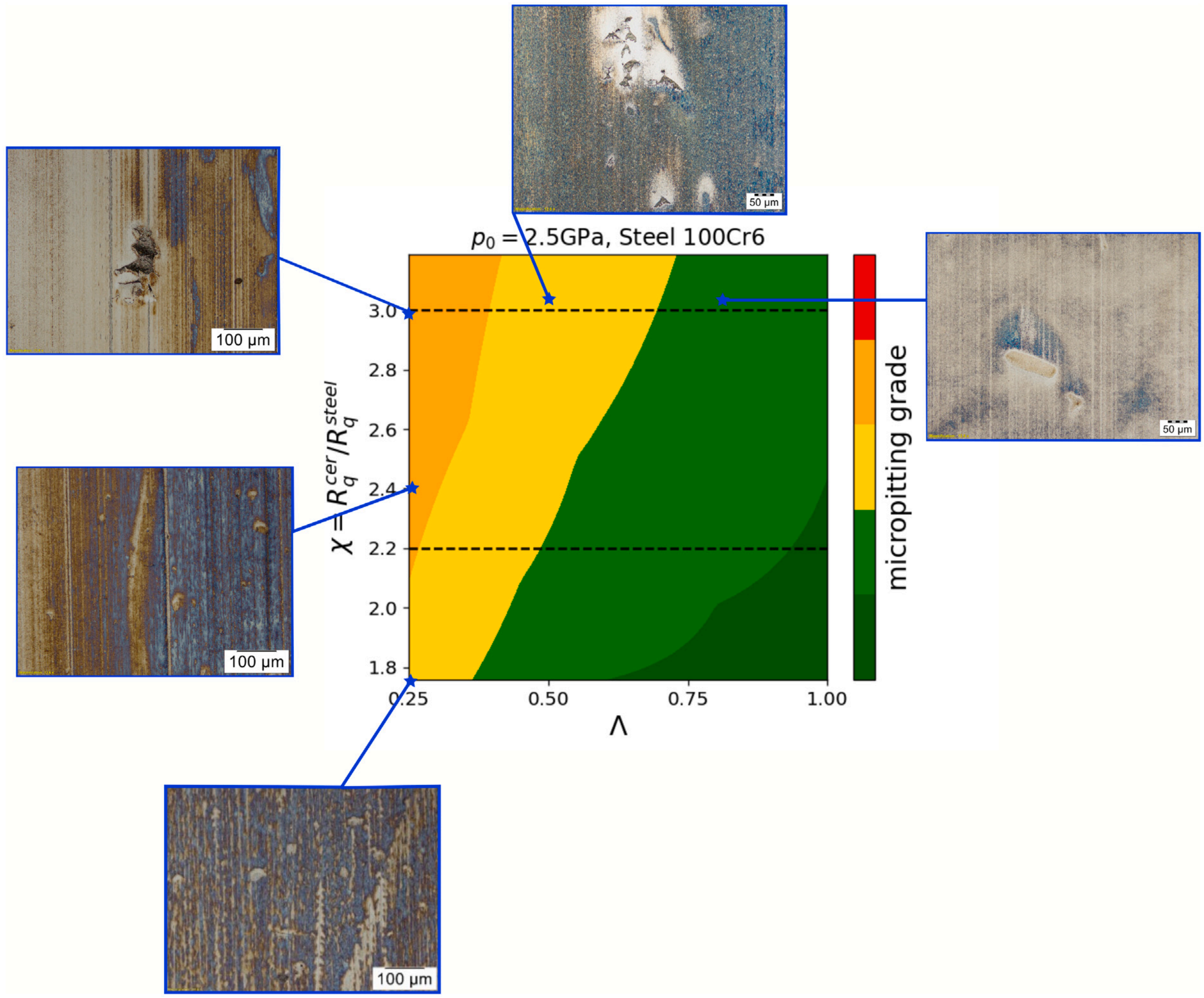

Fig. 14. Predicted performance map as a function of Λ and χ at p_0 =2.5 GPa of 100Cr6 steel versus experimental observation in MPR tests.

Thus, we conclude that hybrid bearings are expected to have superior performance in applications characterized with reduced lubrication conditions.

A new micropitting model (LPMC) based on the energy increase due to localized plasticity has been introduced. The micropitting model, which uses as input basic mechanical properties of the steels, accurately predicts their performance (without free fitting parameters). Furthermore, the new model suggests that steels with high yield strength and reduced work hardening are expected to have a good micropitting performance. Experimental results indicate that the bearing steels tested here are already optimized for good micropitting resistance. However, the theoretical finding presented here can serve as guidance for application-specific steel optimization.

A common method for predicting micropitting in rolling–sliding contact is based on Dang-Van fatigue criterion [2,7,50] and expressed as

$$1 = \max\left\{ \frac{\tau}{\frac{A}{\sqrt{3}\log(N_T)} + \frac{B}{\sqrt{3}} - \beta\sigma_H} \right\} \tag{17}$$

where τ and σ_H are the calculated near-surface shear and hydrostatic stresses; therefore, τ and σ_H depend on the specific contact and lubrication conditions. Furthermore, $\beta = 0.232$ is a material constant, typical for bearing steels calculated using the ratio between shear τ and hydrostatic stress σ_H. The minus sign in front σ_H amplifies the effect of the surface tensile stresses on the accumulated damage since bearing steels are typically more sensitive to the tensile stresses. For more details check Figure 3b in [51]. Finally, A and B are the material constants extracted from the so-called Wöhler curve

$$\sigma_f = A\log(N_T) + B \tag{18}$$

which is obtained in cyclic fatigue tests. In the previous equation, σ_f is the fatigue strength for N_T stress cycles. The material decay after N_T number of cycles is expressed through the parameter A, while the material strength at $N = 1$ is expressed by the parameter B. Dang Van criterion is shown to be a very elegant and computationally simple method since predicting micropitting resistance does not require estimation of the subsurface plastic strain in the rough contact. However, using Dang Van criterion for comparing the micropitting resistance of different steels with different chemical composition, production route and source, . . . should be taken with extra care since measured fatigue properties of steels can be significantly affected by the steel cleanliness (presence of nonmetallic inclusions [52]) and sample roughness. On contrary, micropitting failure mode is generally not affected by the presence of inclusions since (i) their size is typically larger than the size of micropitts, and (ii) micropitting is mostly driven by the rough contact surface stresses.

Due to fractal properties of the functional surfaces, the new micropitting criterion has also implications for predicting *adhesive wear* resistance in partly lubricated sliding contact. Formation of wear particles in sliding contact is typically predicted using Archard law [53] in which particle formation rate (removal rate of asperity) is $\Delta h/\Delta N_t \propto k/H$, where H is material hardness, and k is wear coefficient which can be associated with the wear particle detachment probability.

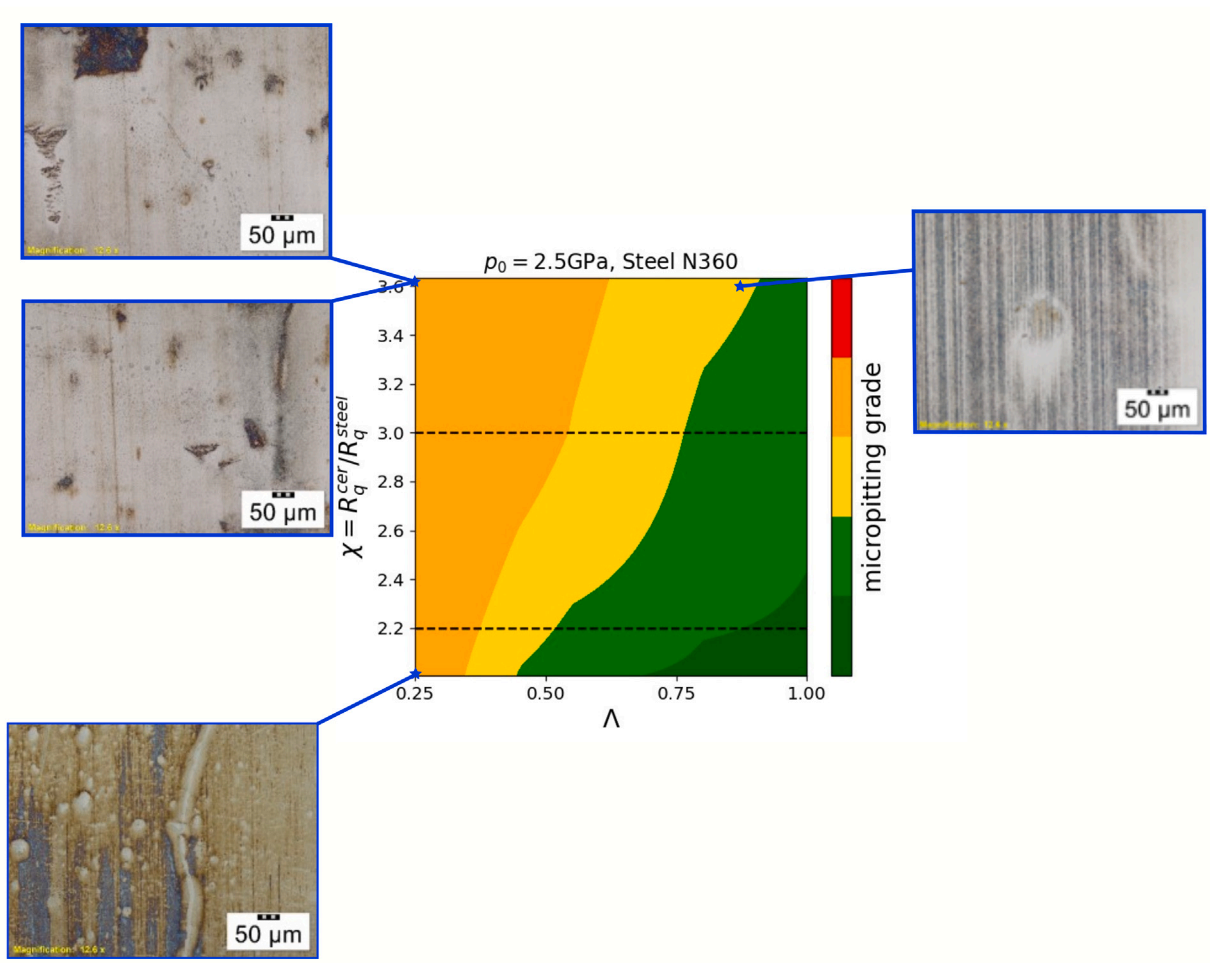

Fig. 15. Predicted performance map as a function of Λ and χ at p_0 =2.5 GPa of N360 steel versus experimental observation in MPR tests.

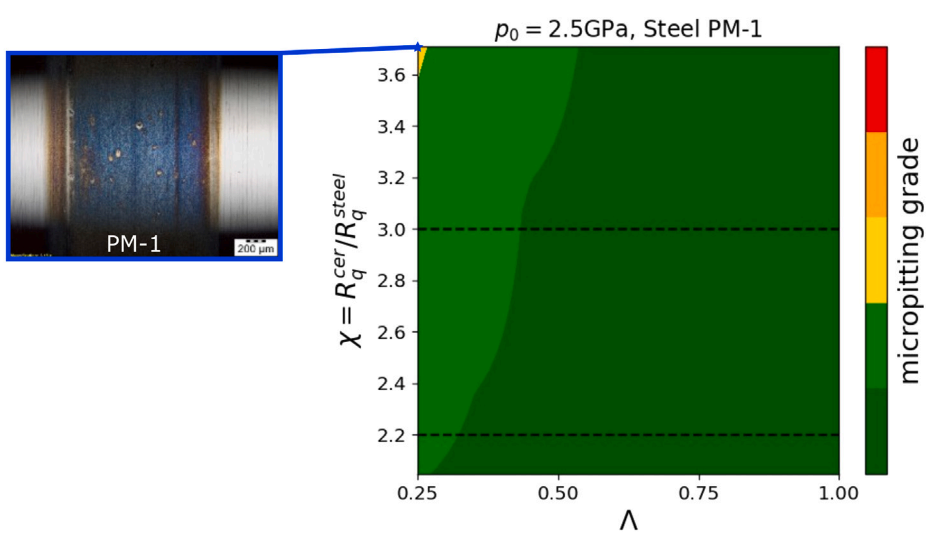

Fig. 16. Predicted performance map as a function of Λ and χ at p_0 =2.5 GPa of PM-1 steel versus experimental observation in MPR tests.

However, even though Archard model is widely used in various tribological applications, the constant k remains an empirical parameter whose values can span several orders of magnitude [54]. More recently, Aghababaei et al. [55] introduced a wear model based on critical length scale which is defined as

$$d^* = 3\frac{G}{\tilde{\tau}^2/2\mu} \tag{19}$$

where G is the material fracture energy and $\tilde{\tau}/2\mu$ is the elastic energy density of an asperity with shear modulus μ when it is loaded to it shear limit $\tilde{\tau}$. [54]. The asperity will detach if its size d is larger than d^*. Since d^* represents a material constant, this model is a first approach for advancement of the wear predictive capabilities as a function of the basic material properties. Eq. (19) suggests that increasing the material fracture energy will lead to improved wear resistance, which is

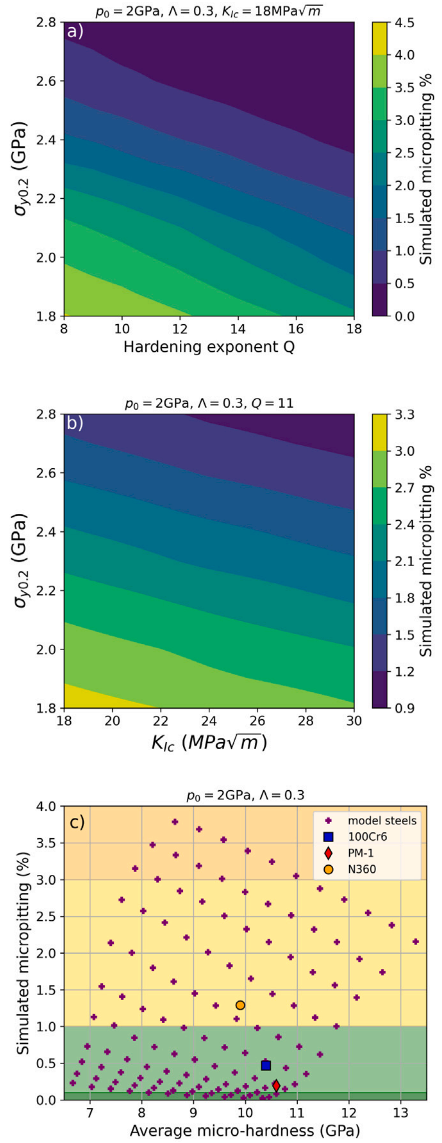

Fig. 17. Simulated percentage of micropitting (a) as a function of the materials hardening exponent Q and yield strength $\sigma_{y0.2}$, $K_{Ic} = 18$ MPa$\sqrt{\text{m}}$; (b) as a function of the materials fracture toughness K_{Ic}, yield strength $\sigma_{y0.2}$ and $Q = 11$; and (c) as a function of the materials average microhardness obtained using combination of different yield strength $\sigma_{y0.2}$, work hardening exponent Q and $K_{Ic} = 18$ MPa$\sqrt{\text{m}}$. The model materials (purple crosses) are compared with actual predictions for 100Cr6 (blue square), PM-1 (red diamond) and N360 (orange circle). The simulated results are obtained for contact conditions corresponding to $p_0 = 2$ GPa and $\Lambda = 0.3$.

generally aligned with experiments. Furthermore, Eq. (19) also suggests that increasing the $\tilde{\tau}$, which is related to the material hardness, will lead to easier formation of wear particles. On contrary, this observation is not aligned with experiments, and potential reasons for this discrepancy have been extensively studied by Molinari and co-workers (e.g. see Ref. [54]).

Comparing the micropitting criterion introduced here and the wear model of Molinari and co-workers, we notice that both criteria are intimately related to the amount of stored energy in the contact zone (asperity) which is available for the nucleation and propagation of a crack. Thus, since the model introduced here is general and not necessarily limited to micropitting, we conclude that by setting $N_c = 1$ and $a = d^*$, and by rearranging Eq. (11) we can derive a new expression for the critical length scale as

$$d^* = \frac{6G}{\mu B \Delta\varepsilon_p^2}. \tag{20}$$

Eq. (20) solves the previous inconsistency since the critical size is inversely proportional to the average plastic strain $\Delta\varepsilon_p^2$ experienced by the asperity. This is the fourth and final result of this manuscript. This relationship clearly suggests that materials which are more resistant to plastic deformation (generally related to higher hardness) will be more resistant to wear, as typically observed in experiments. Nélias et al. [56] proposed a wear model along the similar line in which wear particle will form if the strain within the localized plasticity zone reaches a critical value. However, even though the criteria of Nélias and co-workers is mechanistically robust, estimating the fracture stress and critical strain is not a trivial task.

As in the case of micropitting, $\Delta\varepsilon_p^2$ is also affected by the magnitude of the surface pressure in partly lubricated rough contact; therefore, for a given fracture energy G, the good adhesive resistance is expected for materials with high yield strength $\sigma_{y0.2}$ and reduced capability for work hardening (larger Q). Using the material properties of the three steels tested here, we deduce that PM-1 is expected to be the most resistant steel to adhesive wear. The second performing steels is 100Cr6 martensitic steel, while N360 is expected to be the worst performing steel. Our analytically deduced predictions are in good agreement with the experimental results reported previously by Sherif and co-workers [14]. More quantitative predictions based on Eq. (20) will be reported in near future.

Several extensions of the present analysis are necessary. First, the micropitting performance maps introduced above do not account for the effect of mild wear which has been shown to have beneficial effects on the micropitting resistance [2]. Including the effect of mild wear might reduce the predicted performance difference between 100Cr6 and N360 steels. Second, the onset of material plasticity is thermally activated process at finite temperatures. Therefore, effect of contact temperature on the formation of micropitting needs to be explored. Third, the mechanical properties of the surfaces can be significantly affected by the chemical effects such as tribolayer build up, corrosion effect and/or hydrogen effects. The chemical reactivity of the surfaces is dependent on the steel chemical composition, heat treatment and stress state due to contact between two rough surfaces.

7. Conclusions

The micropitting tests have demonstrated very good micropitting resistance of hybrid contact even with increased roughness levels of the ceramics surfaces. A new micropitting criterion (LPMC) has been shown to accurately predicts ranking of different steel with respect to their micropitting resistance. The model has been extensively used for generating the material performance maps as a function of their basic mechanical properties: (i) yield strength, (ii) work hardening, and (iii) fracture toughness. In addition to the micropitting, the theoretical method introduced here can be applied for predicting the adhesive wear resistance of different steels. More importantly, the model introduced here resolves the inconsistency of the previously introduced wear model and related to the definition of the critical length scale for the formation of wear particle.

CRediT authorship contribution statement

Predrag Andric: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Conceptualization. **Victor Brizmer:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Boris Minov:** Writing – review & editing, Validation, Investigation, Formal analysis. **Tomas Nuijten:** Writing – review & editing, Methodology, Investigation. **Roel van der Zwaan:** Writing – review & editing, Methodology, Investigation. **Guillermo E. Morales-Espejel:** Writing – review & editing, Supervision, Software, Methodology, Investigation, Formal analysis. **Charlotte Vieillard:** Writing – review & editing, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. The plasticity model

Consider that the uniaxial constitutive behavior of the subsurface of an elasto-plastic material can be described using the Ramberg–Osgood (RO) power-law equation [57]

$$\varepsilon = \frac{\sigma}{E} + \left(\frac{\sigma}{B}\right)^{N} \tag{A.1}$$

where $\varepsilon = \varepsilon_e + \varepsilon_p$ is the total strain composed of elastic ε_e and plastic ε_p parts, N is the material hardening exponent, $B = \sigma_{y0.2}/0.002^{(1/K)}$ is the material plasticity modulus, and $\sigma_{y0.2}$ is the material yield strength at 0.2% offset. For a given stress state σ, there are two unknowns to be determined, namely ε_p and ε_e, and one available equation (Eq. (A.1)). An approximate solution for the missing equation, based on the strain energy density equivalence for the "fictitious" elastic state and for the real elastic–plastic one, was proposed by Glinka in the form of [58]

$$\frac{1}{2}\sigma_{el}\varepsilon_{el} = \int_0^{\varepsilon} \sigma d\varepsilon \tag{A.2}$$

where σ_{el} is the equivalent elastic stress (applied stress), ε_{el} is the equivalent elastic strain (if the material is linear elastic). The above approximation resembles a nonlinear elastic problem rather than the history dependent elasto-plastic problem. However, the approximation proposed is shown to be sufficiently accurate if the plastic zone is small and embedded in a linearly elastic material (small scale yielding condition) [58]. Specifically to RCF, this condition is generally met if the plastically deformed subsurface (around the depth of high von Mises stress) is separated from the half-space surface (raceway) by sufficiently thick linearly-elastic material. The maximum contact pressure which satisfy this condition depends on the material constitutive behavior.

The problem is now well defined, with the two analytical equations and two unknowns to be determined. Inserting the infinitesimal form of Eq. (A.1)

$$d\varepsilon = \frac{d\sigma}{E} + \frac{K}{B}\left(\frac{\sigma}{B}\right)^{K-1} d\sigma \tag{A.3}$$

into Eq. (A.2) and integrating, yields the following analytical nonlinear equation

$$\frac{\sigma^2}{2E} + \frac{K}{(K+1)B^K}\sigma^{K+1} - \frac{\sigma_{el}\varepsilon_{el}}{2} = 0 \tag{A.4}$$

for evaluating σ. Inserting the value of σ into Eq. (A.1) leads to the total strain ε, from which we calculate ε_p as

$$\varepsilon_p = \varepsilon - \frac{\sigma_{el}}{E}. \tag{A.5}$$

Extending the above analysis to the Hertzian contact problem is straight forward if for $\sigma_{el}(x, y, z)$ we use the actual value of the equivalent von Mises stress

$$\sigma_{el}(x,y,z) = \sigma_{\mathrm{vM}}(x,y,z) = \sqrt{\frac{3}{2}\sigma'_{ij}(x,y,z)\sigma'_{ij}(x,y,z)} \tag{A.6}$$

where $\sigma'_{ij}(x, y, z)$ is the deviatoric component of the stress tensor. We have replaced the 3D stress problem by an uniaxial one in which the uniaxial stress component corresponds to the equivalent subsurface von Mises stress. The above analysis, even though it can be regarded as a rough approximation, satisfies the conditions of continuity and equilibrium. During one over-rolling in RCF, the plastic strain range can be calculated as

$$\Delta\varepsilon_p = \max(\varepsilon_p). \tag{A.7}$$

References

[1] Olver A. The mechanism of rolling contact fatigue: an update. Proc. Inst Mech Eng, Part J: J Eng Tribol 2005;219(5):313–30.
[2] Morales-Espejel GE, Brizmer V. Micropitting modelling in rolling–sliding contacts: application to rolling bearings. Tribol Trans 2011;54(4):625–43.
[3] Gouda K, Rycerz P, Kadiric A, Morales-Espejel G. Assessing the effectiveness of data-driven time-domain condition indicators in predicting the progression of surface distress under rolling contact. Proc Inst Mech Eng, Part J: J Eng Tribol 2019;233(10):1523–40.
[4] Spikes H, Olver A, Macpherson P. Wear in rolling contacts. Wear 1986;112(2):121–44.
[5] Olver A, Spikes H, Bower A, Johnson K. The residual stress distribution in a plastically deformed model asperity. Wear 1986;107(2):151–74.
[6] Morales-Espejel G, Brizmer V, Stadler K. Understanding and preventing surface distress. Gear Solut July 2012;29:38.
[7] Brizmer V, Pasaribu H, Morales-Espejel GE. Micropitting performance of oil additives in lubricated rolling contacts. Tribol Trans 2013;56(5):739–48.
[8] Morales-Espejel G, Rycerz P, Kadiric A. Prediction of micropitting damage in gear teeth contacts considering the concurrent effects of surface fatigue and mild wear. Wear 2018;398:99–115.
[9] Rycerz P, Kadiric A. The influence of slide–roll ratio on the extent of micropitting damage in rolling–sliding contacts pertinent to gear applications. Tribol Lett 2019;67:1–20.
[10] Ueda M, Spikes H, Kadiric A. In-situ observations of the effect of the ZDDP tribofilm growth on micropitting. Tribol Int 2019;138:342–52.
[11] Mallipeddi D, Norell M, Naidu VS, Zhang X, Näslund M, Nyborg L. Micropitting and microstructural evolution during gear testing-from initial cycles to failure. Tribol Int 2021;156:106820.
[12] Ueda M, Wainwright B, Spikes H, Kadiric A. The effect of friction on micropitting. Wear 2022;488:204130.
[13] Wainwright B, Takeuchi H, Makino T, Kadiric A. The influence of Λ ratio and surface roughness on the initiation and progression of micropitting damage. Wear 2022;508:204473.
[14] Sherif MY, Brizmer V, Meeuwenoord R, Matta C, Broitman E, Nuijten T. The influence of steel microstructure in high-speed high-load bearing applications. Mater Sci Technol 2021;37(17):1370–85.
[15] Morales-Espejel GE, Brizmer V. Effect of materials and surfaces on frictional heating resistance of high-speed high-load rolling bearings. Proc Inst Mech Eng, Part J: J Eng Tribol 2022;236(4):583–94.
[16] Oila A, Shaw B, Aylott C, Bull S. Martensite decay in micropitted gears. Proc Inst Mech Eng, Part J: J Eng Tribol 2005;219(2):77–83.
[17] Wan G, Gabelli A, Ioannides E. Increased performance of hybrid bearings with silicon nitride balls. Tribol Trans 1997;40(4):701–7.
[18] Morales-Espejel G, Gabelli A, Vieillard C. Hybrid bearings lubricated with pure refrigerant liquid. Turbomachinery 2004;32(7):387–94.
[19] Brizmer V, Gabelli A, Vieillard C, Morales-Espejel GE. An experimental and theoretical study of hybrid bearing micropitting performance under reduced lubrication. Tribol Trans 2015;58(5):829–35.
[20] Beswick J. Fracture and fatigue crack propagation properties of hardened 52100 steel. Metall Trans A 1989;20:1961–73.
[21] Vieillard C, Kadin Y, Morales-Espejel GE, Gabelli A. An experimental and theoretical study of surface rolling contact fatigue damage progression in hybrid bearings with artificial dents. Wear 2016;364:211–23.
[22] Morales-Espejel GE, Wemekamp A, Félix-Quiñonez A. Micro-geometry effects on the sliding friction transition in elastohydrodynamic lubrication. Proc Inst Mech Eng, Part J: J Eng Tribol 2010;224(7):621–37.
[23] Stanley H, Kato T. An FFT-based method for rough surface contact. 1997.
[24] Johnson KL, Johnson KL. Contact mechanics. Cambridge University Press; 1987.
[25] Frérot L, Bonnet M, Molinari J-F, Anciaux G. A Fourier-accelerated volume integral method for elastoplastic contact. Comput Methods Appl Mech Engrg 2019;351:951–76.
[26] Tabor D. The area of contact between stationary and between moving surfaces (Ph.D. thesis), University of Cambridge; 1939.
[27] Tabor D. The hardness of metals. Oxford University Press; 2000.
[28] Tian X, Bhushan B. A numerical three-dimensional model for the contact of rough surfaces by variational principle. 1996.
[29] Almqvist A, Sahlin F, Larsson R, Glavatskih S. On the dry elasto-plastic contact of nominally flat surfaces. Tribol Int 2007;40(4):574–9.

[30] Brizmer V, Kligerman Y, Etsion I. The effect of contact conditions and material properties on the elasticity terminus of a spherical contact. Int J Solids Struct 2006;43(18–19):5736–49.

[31] George EP, Curtin W, Tasan CC. High entropy alloys: A focused review of mechanical properties and deformation mechanisms. Acta Mater 2020;188:435–74.

[32] Gao Y, Bower A, Kim K-S, Lev L, Cheng Y. The behavior of an elastic–perfectly plastic sinusoidal surface under contact loading. Wear 2006;261(2):145–54.

[33] Ghaednia H, Wang X, Saha S, Xu Y, Sharma A, Jackson RL. A review of elastic–plastic contact mechanics. Appl Mech Rev 2017;69(6):060804.

[34] Pei L, Hyun S, Molinari J-F, Robbins MO. Finite element modeling of elasto-plastic contact between rough surfaces. J Mech Phys Solids 2005;53(11):2385–409.

[35] ASTM E, et al. Standard practices for cycle counting in fatigue analysis. 2011.

[36] Lai J, Lund T, Rydén K, Gabelli A, Strandell I. The fatigue limit of bearing steels–Part I: A pragmatic approach to predict very high cycle fatigue strength. Int J Fatigue 2012;38:155–68.

[37] Gabelli A, Lai J, Lund T, Rydén K, Strandell I, Morales-Espejel GE. The fatigue limit of bearing steels–Part II: Characterization for life rating standards. Int J Fatigue 2012;38:169–80.

[38] Kato M, Onaka S, Mori T, Mura T. Statistical consideration of plastic strain accumulation in cyclic deformation and fatigue crack initiation. Tech. rep., Northwestern Univ Evanston IL; 1985.

[39] Eshelby JD. The elastic field outside an ellipsoidal inclusion. Proc R Soc Lond Ser A Math Phys Eng Sci 1959;252(1271):561–9.

[40] Mura T. Micromechanics of defects in solids. Springer Science & Business Media; 2013.

[41] Mura T. Micromechanics of defects in solids. Springer Science & Business Media; 2013.

[42] Molski K, Glinka G. A method of elastic-plastic stress and strain calculation at a notch root. Mater Sci Eng 1981;50(1):93–100.

[43] Andric P, Restrepo SE, Maresca F. Predicting dislocation density in martensite ab-initio. Acta Mater 2023;243:118500.

[44] Ju J, Sun L. A novel formulation for the exterior-point eshelby's tensor of an ellipsoidal inclusion. J Appl Mech 1999;66(2):570–4.

[45] Lemaitre J, Desmorat R. Engineering damage mechanics: ductile, creep, fatigue and brittle failures. Springer Science & Business Media; 2005.

[46] Lemaitre J. A course on damage mechanics. Springer Science & Business Media; 2012.

[47] Andric P, Restrepo SE, Lai J, Venner CH, Vegter E. Mechanistic driving force for martensite decay in rolling contact fatigue. Tribol Int 2023;184:108470.

[48] Frérot L, Anciaux G, Molinari J-F. Crack nucleation in the adhesive wear of an elastic-plastic half-space. J Mech Phys Solids 2020;145:104100.

[49] Tvergaard V, Hutchinson JW. The relation between crack growth resistance and fracture process parameters in elastic-plastic solids. J Mech Phys Solids 1992;40(6):1377–97.

[50] Dang-Van K. Macro-micro approach in high-cycle multiaxial fatigue. In: Advances in multiaxial fatigue. ASTM International; 1993.

[51] Alfredsson B, Arregui IL, Lai J. Low temperature creep in a high strength roller bearing steel. Mech Mater 2016;100:109–25.

[52] Oezel M, Janitzky T, Beiss P, Broeckmann C. Influence of steel cleanliness and heat treatment conditions on rolling contact fatigue of 100Cr6. Wear 2019;430:272–9.

[53] Archard J. Contact and rubbing of flat surfaces. J Appl Phys 1953;24(8):981–8.

[54] Brink T, Frérot L, Molinari J-F. A parameter-free mechanistic model of the adhesive wear process of rough surfaces in sliding contact. J Mech Phys Solids 2021;147:104238.

[55] Aghababaei R, Warner DH, Molinari J-F. Critical length scale controls adhesive wear mechanisms. Nature Commun 2016;7(1):11816.

[56] Nelias D, Boucly V, Brunet M. Elastic-plastic contact between rough surfaces: proposal for a wear or running-in model. 2006.

[57] Ramberg W, Osgood WR. Description of stress-strain curves by three parameters. Tech. rep., 1943.

[58] Glinka G. Energy density approach to calculation of inelastic strain-stress near notches and cracks. Eng Fract Mech 1985;22(3):485–508.